Towards Reliable VLM Judges: State-Conditional Invariance and Presentation-Aware Diagnostics

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Abstract

Vision-language models (VLMs) are increasingly used as automated judges for multimodal and embodied tasks, yet aggregate accuracy or consistency alone does not reveal whether a judge changes its verdict for the right reason. We propose State-Conditional Invariance (SCI), a reliability principle requiring a judge to remain stable under semantics-preserving changes in presentation while responding to causal interventions that change the task outcome. We instantiate SCI in **GridWM-Judge**, a simulator-grounded diagnostic benchmark built from deterministic MiniGrid trajectories with aligned Full, evidence-removal, and counterfactual variants, plus presentation and representation probes. Across five diagnostic tasks, GridWM-Judge tests local perception, action-conditioned transitions, presentation stability, representation exchangeability, and Full–CF causal discrimination. Evaluating current VLM judges reveals separable failures, including presentation sensitivity, representation fragility, causal misgrounding, action-insensitive heuristics, and stability traps. These results show that reliable VLM-as-a-judge evaluation requires joint reporting of correctness, presentation invariance, and causal sensitivity rather than a single accuracy or consistency score.

1 Introduction

Vision-language models (VLMs) are increasingly used not only as answer generators but also as automated judges. In text-only settings, LLM-based judges have been used to compare conversational responses and evaluate instruction-following systems [1–3]. In multimodal settings, this paradigm has expanded to rubric-based vision-language evaluation and in-the-wild preference assessment [4–6], as well as multimodal reward, critic, and agent evaluation [7–9]. Such judges are attractive because they provide scalable alternatives to rule-based metrics and expensive human annotations. However, once a VLM judge is used to decide whether an embodied trajectory succeeds or fails, reliability becomes more subtle than aggregate accuracy: a judge may be correct for the wrong reason, stable because it defaults to one verdict, or unstable under changes that should not affect the task outcome.

This raises a simple but fundamental question: *when should a vision-language judge change its mind?* Our answer is that a reliable judge should change its verdict when the underlying world changes, but not when only the presentation of the same evidence changes. For example, a successful trajectory should receive the same verdict under different layouts, explicitly indexed frame orderings, rendering styles, or prompt framing; conversely, a minimal intervention that changes the trajectory outcome from success to failure should change the verdict. We formalize this requirement as *State-Conditional Invariance* (SCI): given the same simulator-grounded trajectory semantics s, a judge's verdict $\hat{y}$ should be conditionally independent of nuisance presentation variables ϕ,

$$\hat{y} \perp \phi \mid s. \tag{1}$$

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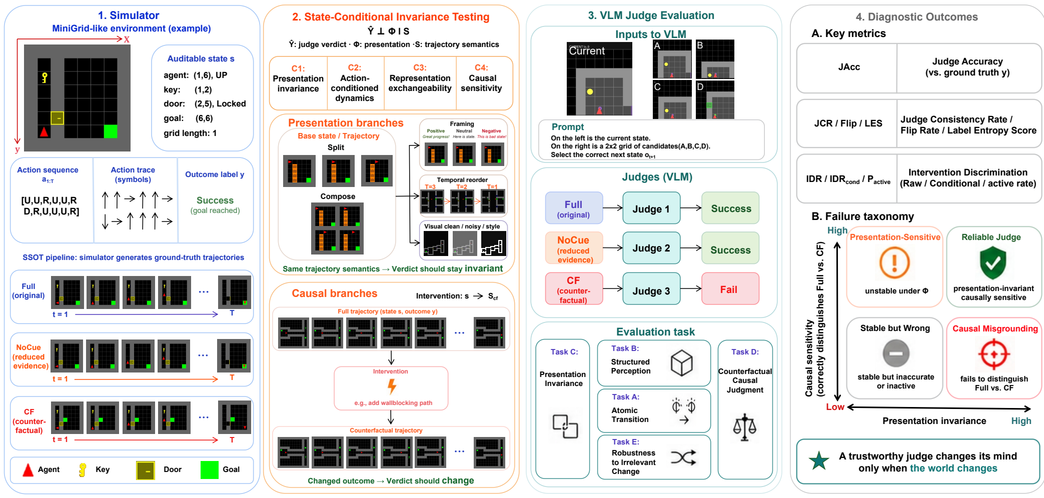


Figure 1: **Overview of GridWM-Judge.** The simulator provides auditable state-action trajectories and outcome labels. SCI separates presentation changes, where the same trajectory semantics should yield the same verdict, from causal interventions, where a changed outcome should change the verdict. The resulting tasks and metrics diagnose presentation sensitivity, representation fragility, causal misgrounding, action-insensitive heuristics, and stability traps.

SCI is therefore a selective invariance principle: the judge should be insensitive to how evidence is presented, but sensitive to what actually happened. We use "presentation" broadly to include prompt framing, visual perturbations, temporal ordering, and evidence interfaces. In the diagnostic decomposition below, C1 focuses on temporal, visual, and framing probes, while C3 isolates representation exchangeability between composite and split-image evidence formats.

Existing multimodal judge benchmarks do not fully isolate this distinction. Many evaluate agreement with human preferences, rubric-following ability, reward-model quality, or arena-style rankings [5–8]. These evaluations are valuable, but they often conflate two questions: whether a judge is robust to nuisance presentation changes, and whether it remains responsive to causal changes in the underlying task outcome. This distinction is especially important for embodied evaluation, where a counterfactual trajectory may look visually similar to a successful one while differing in a causally decisive event. In other words, existing benchmarks either evaluate judge agreement, preference alignment, or embodied reasoning ability, but rarely provide paired controls that jointly separate presentation invariance from causal sensitivity.

We introduce **GridWM-Judge**, a grid-world and world-model-inspired diagnostic benchmark for evaluating VLM judges under SCI. Built on deterministic grid-world environments, GridWM-Judge uses the simulator as a Single Source of Truth (SSOT) for states, actions, events, and success predicates. For each trajectory group, it constructs aligned FULL, evidence-removal, and counterfactual variants, and evaluates them under presentation probes including composite and split-image representations. This design separates presentation variants, where the world is fixed but the evidence interface changes, from causal counterfactuals, where an intervention changes the task outcome.

Figure 1 summarizes the overall diagnostic pipeline, from simulator-grounded trajectory generation to presentation and causal tests under SCI, and maps the resulting tasks to the failure modes analyzed in this paper. Our main contributions can be summarized as:

- We formulate VLM judge reliability as *State-Conditional Invariance*, separating nuisance presentation changes from causal outcome changes.
- We instantiate SCI in GridWM-Judge, an executable benchmark with aligned trajectory variants, counterfactual interventions, and presentation probes.
- We diagnose current VLM judges and identify presentation leakage, representation fragility, causal misgrounding, action-insensitive heuristics, and stability traps.

2 Related Work

Model-based judges and multimodal evaluation. LLM-as-a-judge is now widely used for scalable evaluation of open-ended model outputs, from comparative or rubric-based text evaluation [1–3, 10–12] to multimodal judging, preference assessment, reward modeling, and critic or agent evaluation [4–8, 13, 14, 9]. These works establish the utility of model-based judges, but primarily study human preference alignment, rubric following, or reward quality. GridWM-Judge instead asks whether a VLM judge changes its verdict for the right reason under simulator-grounded presentation and causal controls.

Judge reliability, bias, and presentation effects. A growing literature studies the reliability of model judges themselves, including fairness concerns, selection bias, verbosity bias, rubric sensitivity, and inconsistency across prompts or candidates [15–18]. In multimodal judging, VLM evaluators may be unreliable under benign changes in video evidence, judge composition, or input presentation [19, 20]. Separately, VLMs are known to suffer from visual hallucination, image-context reasoning failures, and object hallucination [21–23]. Long-context and image-sequence studies further show that models can depend on where and how evidence is presented [24, 25]. These findings motivate treating the judge itself as the object of evaluation. Our work operationalizes this idea by using a simulator-backed source of truth to separate presentation leakage, representation fragility, causal misgrounding, and degenerate stability.

Embodied evaluation, world models, and counterfactual diagnostics. Controlled environments such as BabyAI and MiniGrid support auditable grounded language and navigation experiments [26, 27], while video and embodied benchmarks such as CLEVRER and COVER evaluate causal or counterfactual reasoning over dynamic scenes [28, 29]. World-model research studies action-conditioned prediction, planning, and generative environment modeling [30–38]. Recent grid-world and VLM diagnostics ask whether models solve, plan, act, or form internal world-model-like representations [39–42]; GridWM-Judge instead evaluates whether a VLM can reliably judge what happened in a trajectory under aligned presentation and causal counterfactuals.

3 State-Conditional Invariance Framework

We formalize VLM judge reliability as *State-Conditional Invariance* (SCI). Our goal is not to infer whether a VLM has an internal world model, but to test whether its success/failure verdict behaves correctly under controlled changes to the world and to the presentation of evidence. A reliable judge should remain invariant under semantics-preserving presentation changes, while remaining sensitive to causal interventions that change the task outcome. This selective-invariance view is related to invariant prediction, causal invariance, counterfactual reasoning, and shortcut learning [43–46], but we apply it to judge verdicts rather than to learned predictors, fairness criteria, or agent policies.

3.1 Setup and SCI Principle

Let $g \in \mathcal{G}$ denote a simulator-generated trajectory group. Each group contains an aligned state trajectory, action sequence, rendered evidence, mission text, outcome label, and audit metadata. We write

$$s_g = (s_{g,0:T}, a_{g,0:T-1}, m_g), \qquad \ell_g = h(s_g), \qquad \hat{y}_g^{v,\phi} \in \{0,1\}, \tag{2}$$

where s_g denotes simulator-grounded trajectory semantics, h is the simulator success predicate, $\ell_g \in \{0,1\}$ is the resulting success/failure label, v is the trajectory variant, and ϕ is the presentation variable. In our benchmark, $v \in \{\mathrm{full}, \mathrm{nocue}, \mathrm{cf}\}$ denotes a successful full trajectory, an evidence-removal diagnostic variant, and a counterfactual failure variant, respectively. The presentation variable ϕ can encode prompt framing, temporal order, visual rendering style, or representation format.

SCI requires that, conditioned on the same underlying trajectory semantics, the judge's verdict should be independent of nuisance presentation variables:

$$\hat{Y} \perp \Phi \mid S. \tag{3}$$

Equivalently, the judge should remain stable when only presentation changes, but should change when a causal intervention changes the outcome:

$$\hat{y}_g^{\mathrm{full},\phi} = \hat{y}_g^{\mathrm{full},\phi_0} \quad \text{for } \phi \in \Phi_{\mathrm{eq}}(s_g), \qquad \hat{y}_g^{\mathrm{full},\phi_0} = 1,\ \hat{y}_g^{\mathrm{cf},\phi_0} = 0. \tag{4}$$

Table 1: **SCI diagnostic conditions and their benchmark instantiations.** NS denotes nuisance stability, JCR denotes judge consistency rate, LES denotes leakage effect size, VCC denotes verdict consistency coefficient, and IDR denotes intervention discrimination rate. Full mathematical definitions of these variables and metrics are provided in Appendix A.

Condition	Question	Instantiation	Metrics
C1: Presentation invariance	Does the verdict stay stable under semantics-preserving presentation changes?	Task C temporal, visual, and framing probes	NS, JCR, LES
C2: Action-conditioned dynamics	Does the model use action-conditioned transitions rather than superficial visual matching?	Task A next-observation prediction with action shuffle/mask ablations	Acc_{A1}, $\Delta_{\mathrm{shuffle}}$, Δ_{mask}
C3: Representation exchangeability	Does the verdict remain stable across equivalent evidence formats?	Composite vs. split-image representations; optional oracle state tables	$\mathrm{VCC}_{\mathrm{rep}}$
C4: Causal sensitivity	Does the verdict change when a causal intervention changes the outcome?	Task D paired FULL vs. CF trajectories	p_{active}, $\mathrm{IDR}_{\mathrm{raw}}$, $\mathrm{IDR}_{\mathrm{cond}}$

Thus, SCI is not ordinary robustness: it asks whether invariance is appropriate. A reliable judge should ignore nuisance presentation changes, but respond to state changes that alter the task outcome.

3.2 Diagnostic Conditions and Metrics

GridWM-Judge instantiates SCI through four diagnostic conditions. Table 1 summarizes the reliability question, benchmark instantiation, and main metrics. The tasks are not independent leaderboards; they diagnose complementary failure sources in the same judge.

We report four classes of metrics. Correctness anchors, such as $\mathrm{JAcc}_{\mathrm{neu}}$ and $\mathrm{Acc}_{\mathrm{cf}}$, measure whether the judge accepts successful trajectories and rejects counterfactual failures. Presentation and representation metrics, such as NS, JCR, LES, and $\mathrm{VCC}_{\mathrm{rep}}$, measure whether verdicts remain stable under semantics-preserving changes. Causal-sensitivity metrics, especially p_{active}, $\mathrm{IDR}_{\mathrm{raw}}$, and $\mathrm{IDR}_{\mathrm{cond}}$, measure whether the judge discriminates successful full trajectories from matched counterfactual failures. Task-specific controls, including Task A/B/E metrics, help distinguish perception or action-conditioning failures from higher-level causal judgment failures.

We interpret consistency metrics only together with correctness and causal sensitivity. A model can be stable simply because it always predicts the same label. We therefore summarize failures as presentation leakage, representation fragility, action-ignorant heuristics, causal misgrounding, and stability traps; full definitions and degenerate-judge criteria are provided in Appendix A.

4 GridWM-Judge: Benchmark and Data Construction

GridWM-Judge instantiates the SCI framework in executable grid-world tasks. The benchmark is designed around three requirements: (i) the simulator provides an auditable source of truth for states, actions, events, and outcomes; (ii) each item supports controlled presentation changes and causal interventions; and (iii) the tasks diagnose complementary components of judge reliability rather than forming a single leaderboard. Figure 2 summarizes the benchmark anatomy, task-specific source units, and the boundary between model-facing and audit-only data.

4.1 Environment Suite

We use six deterministic MiniGrid environments as the Tier-0 evaluation suite [27]. They cover object interaction, ordered door opening, hazard navigation, memory, and multi-room navigation, while remaining compact enough to provide symbolic states, rendered observations, discrete actions, and unambiguous success predicates. MiniGrid is not intended to approximate open-ended embodied benchmarks [47]; instead, it lets us audit whether a VLM judge changes its verdict when the world changes or merely when the same evidence is presented differently. Full environment cards are provided in Appendix B.

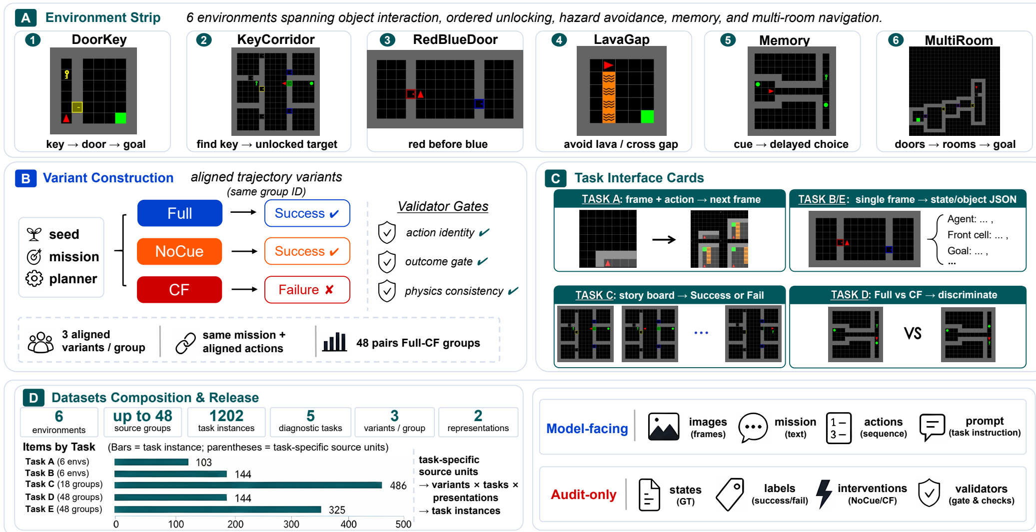


Figure 2: **Benchmark anatomy and dataset composition.** GridWM-Judge uses task-specific source units: trajectory-level judging tasks are built from aligned trajectory groups, while local perception and transition-control tasks are generated from environment-level states or local transitions. These source units yield model-facing task instances after expanding variants, probes, and representation across five diagnostic tasks, with model-facing inputs separated from judge-only audit metadata.

4.2 Trajectory Groups and Variants

For trajectory-level judging tasks, the source unit is a trajectory group $g \in \mathcal{G}$, which contains aligned simulator states, actions, rendered frames, mission text, outcome labels, and audit metadata. For each sampled environment seed, we first construct a successful FULL trajectory using a deterministic planner. We then derive matched variants when supported by the task: NOCUE, which removes selected early evidence while preserving the success outcome, and CF, a counterfactual failure generated by a minimal simulator intervention followed by rerollout under the unchanged action sequence. Thus, NOCUE probes evidence dependence, whereas CF tests whether the judge responds to a causal outcome change. We do not treat NOCUE as a semantics-preserving presentation transform; it removes selected evidence while preserving the outcome and is used only as an evidence-dependence diagnostic.

Local perception and transition-control tasks are generated from task-specific source units, such as environment-level states or local transitions, and are reported as model-facing task instances. This distinction lets GridWM-Judge combine trajectory-level Full/NoCue/CF diagnostics with local Task A/B/E controls without treating all items as the same type of group. All generated items pass task-appropriate validation gates before entering the benchmark, including action-identity checks across aligned variants, FULL-success and CF-failure outcome gates, masking-window constraints, interaction preservation checks, physics consistency checks, and symbolic state-encoding checks when applicable. Full generation details and validator definitions are given in Appendix B.

4.3 Presentations and Diagnostic Tasks

GridWM-Judge treats presentation as a controlled experimental variable. For the same trajectory group, the evidence may be shown as a composite storyboard or as split images with clearer per-frame evidence; we also include temporal, visual, and prompt-framing probes. These presentation variants support C1/C3 diagnostics by testing whether verdicts remain stable when the underlying trajectory semantics are unchanged.

The released benchmark contains five diagnostic tasks. Tasks B and E are perception controls that test whether the model can parse local states, objects, and structured scene information before higher-level judgment failures are interpreted. Task C asks for success/failure judgments under presentation and representation changes. Task A tests action-conditioned transition understanding through next-observation prediction and action ablations. Task D is the primary causal-sensitivity test, comparing successful FULL trajectories with matched CF failures. Full task interfaces, prompt templates, schemas, and representative examples are provided in Appendix C.

4.4 Dataset Release and 3D Extension

The benchmark is released as machine-readable exam files with image paths, prompts, labels, and group metadata. Model-facing fields include only rendered evidence, mission text, action text, and task instructions; audit-only metadata, such as symbolic states, intervention records, labels, rewards, and validation logs, are reserved for auditing and scoring. Full schemas, data cards, metadata, and reproducibility instructions are provided in Appendix F.

Availability. Code, data, prompt templates, validation scripts, scoring scripts, and metadata are provided through anonymized review links: code repository and dataset preview. Full release, schema, license, and reproducibility details are provided in Appendix F. We also include a small MiniWorld extension as a 3D external-validity probe and report it separately in Appendix E, without mixing it with the Tier-0 MiniGrid metrics.

5 Diagnostic Experiments

We evaluate whether current VLM judges satisfy the SCI conditions in Section 3. The main text focuses on three diagnostic questions: whether judges respond to causal interventions, remain stable under semantics-preserving presentation changes, and change behavior across equivalent input representations. Full per-task, per-environment, statistical, and MiniWorld results are reported in Appendix D and Appendix E.

5.1 Setup

Models, tasks, and representations. We evaluate a diverse set of proprietary and open VLM judges, including frontier general-purpose models, lightweight flash-style models, and representative open VLM families when complete runs are available [48–50]. Each model is prompted as an external judge of a trajectory or local transition rather than as an embodied agent. We evaluate the five MiniGrid diagnostic tasks in Section 4: Tasks B/E as perception controls, Task C for presentation and representation sensitivity, Task A for action-conditioned transition understanding, and Task D for causal sensitivity. The key representation comparison is between composite storyboards and split-image inputs, which encode the same trajectory semantics but expose evidence through different interfaces.

Inference, metrics, and statistics. All models receive task-specific instructions with the same model-facing fields: rendered evidence, mission text when available, action text, and the required answer format, following standard instruction-following and judge-prompting protocols [51, 1, 2]. We use deterministic decoding when supported and count invalid, missing, or unparsable outputs as incorrect. The main Task D metrics are $\mathrm{JAcc}_{\mathrm{neu}}$, $\mathrm{Acc}_{\mathrm{cf}}$, p_{active}, $\mathrm{IDR}_{\mathrm{raw}}$, and $\mathrm{IDR}_{\mathrm{cond}}$; Task C additionally reports presentation and representation metrics such as NS, JCR, LES, and $\mathrm{VCC}_{\mathrm{rep}}$. We interpret consistency metrics only together with correctness and causal sensitivity, because a constant-verdict judge can be stable without being reliable. Full model identifiers, prompts, parsing rules, bootstrap confidence intervals, and paired significance tests are provided in Appendix F and Appendix D.

5.2 Analysis

Table 2 and Figure 3 summarize the main SCI-facing diagnostic results. We use $(p_{\mathrm{active}}, \mathrm{IDR}_{\mathrm{raw}})$ as the primary causal-sensitivity coordinates because they jointly indicate whether a judge accepts genuine successful trajectories and rejects matched counterfactual failures.

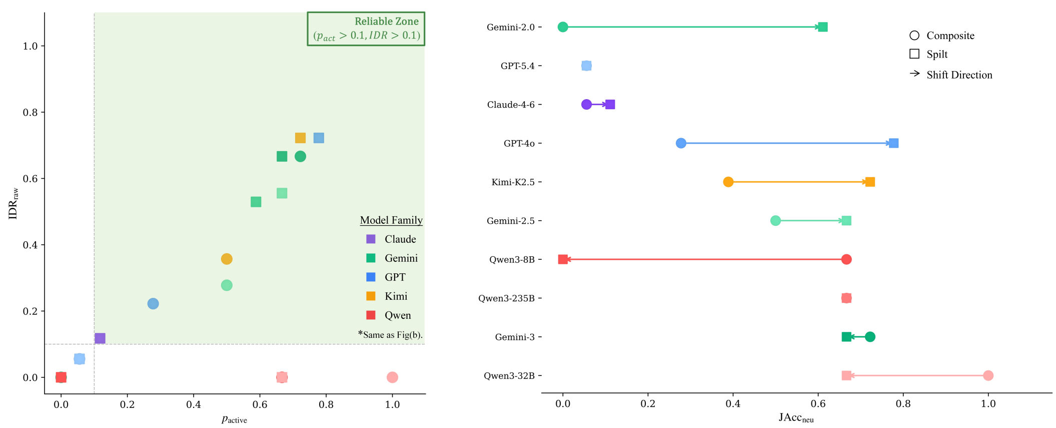
Finding 1: Accuracy does not imply causal sensitivity. Task D tests whether a judge distinguishes successful FULL trajectories from matched counterfactual failures. The key pattern is that $\mathrm{JAcc}_{\mathrm{neu}}$, $\mathrm{Acc}_{\mathrm{cf}}$, and IDR measure different behaviors. A model can obtain high counterfactual accuracy

Table 2: **Main diagnostic results across VLM judges.** C/S denotes Composite/Split representation. Deg. denotes the degenerate status under the corresponding representation; AF: all-fail degenerate, SB: success-biased, —: non-degenerate. Higher p_{active} and IDR_{raw} indicate better causal sensitivity.

Model	JAcc_{neu}	p_{active}		IDR_{raw}		VCC_{rep}	**Deg. (C/S)**
	C/S	**C**	**S**	**C**	**S**		
Claude Sonnet 4.6†	0.06/0.11	0.06	0.12	0.06	0.12	0.924	AF/—
GPT-4o	0.28/0.78	0.28	**0.78**	0.22	**0.72**	0.706	—/—
GPT-5.4	0.06/0.06	0.06	0.06	0.06	0.06	**1.000**	AF/AF
Gemini 3 Flash	0.72/0.67	0.72	0.67	0.67	0.67	0.707	—/—
Gemini 2.5 Flash Lite	0.50/0.67	0.50	0.67	0.28	0.56	0.796	—/—
Gemini 2.0 Flash†	0.00/0.61	0.00	0.59	0.00	0.53	0.544	AF/—
Kimi K2.5	0.39/0.72	0.50	0.72	0.36	**0.72**	0.661	—/—
Qwen3-32B‡	1.00/0.67	1.00	0.67	0.00	0.00	0.667	SB/—
Qwen3-235B	0.67/0.67	0.67	0.67	0.00	0.00	0.743	—/—
Qwen3-8B	0.67/0.00	0.67	0.00	0.00	0.00	0.407	—/AF

† Composite is all-fail degenerate, while Split is non-degenerate.

‡ Success-biased degenerate under Composite.



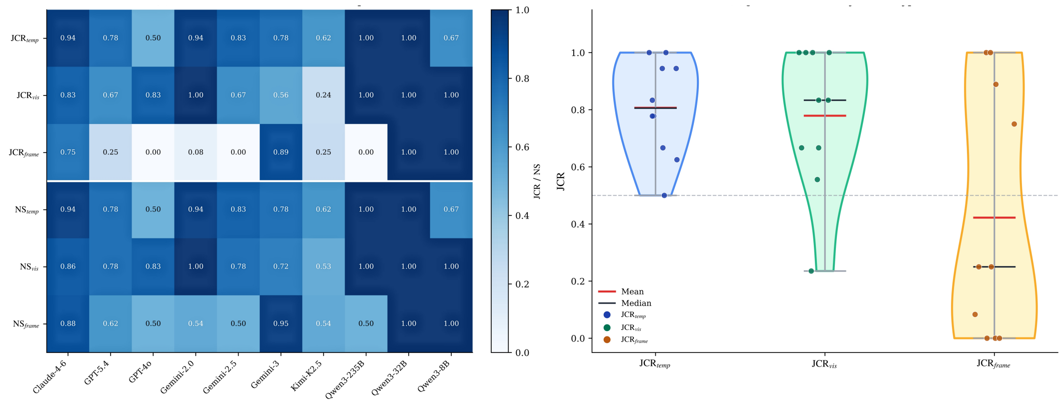
(a) Causal sensitivity phase plot.

(b) Composite-to-split representation shift.

Figure 3: **Main diagnostic results.** (a) plots p_{active} against IDR_{raw}; the ideal region is the upper-right area, where a judge accepts genuine Full successes and rejects matched CF failures. (b) shows how changing the evidence interface from composite to split images shifts correctness and intervention discrimination. A reliable judge should be both causally sensitive and representation-stable.

by predicting failure almost everywhere, yielding low p_{active} and little evidence of intervention discrimination. Conversely, a model can accept successful trajectories while still failing to reject their counterfactual variants. For example, a success-biased judge can accept nearly all FULL trajectories yet obtain zero raw intervention discrimination, illustrating why full-trajectory accuracy alone is insufficient. We therefore interpret causal reliability through $(p_{\text{active}}, \text{IDR}_{\text{raw}})$, with IDR_{cond} used only when the active set is sufficiently large.

Finding 2: Representation changes judge behavior. Task C and the composite–split comparison test whether verdicts remain stable when the underlying trajectory semantics are unchanged. Figure 4 shows that stability differs across temporal, visual, framing, and representation probe families, while Table 3 quantifies the composite-to-split shift. Consistent with prior work on prompt, context, and multimodal evidence sensitivity [2, 24, 25], we find that presenting the same trajectory as split images rather than a composite storyboard can change judging behavior, including full-trajectory acceptance and intervention discrimination. This descriptive improvement should not be read as full reliability: if equivalent representations yield different verdicts, the model violates representation exchangeability. We therefore use VCC_{rep} to quantify whether verdicts are preserved across representations.



(a) Model-by-probe stability.

(b) Probe-family summary.

Figure 4: **Presentation robustness across probe families.** (a) summarizes verdict stability for each model across temporal, visual, framing, and representation probes. (b) aggregates stability across probe families. Higher values indicate greater verdict stability under the corresponding probe family, whereas lower values indicate presentation-induced flips. Representation consistency is reported separately because equivalent evidence formats can yield different verdicts even when trajectory semantics are unchanged.

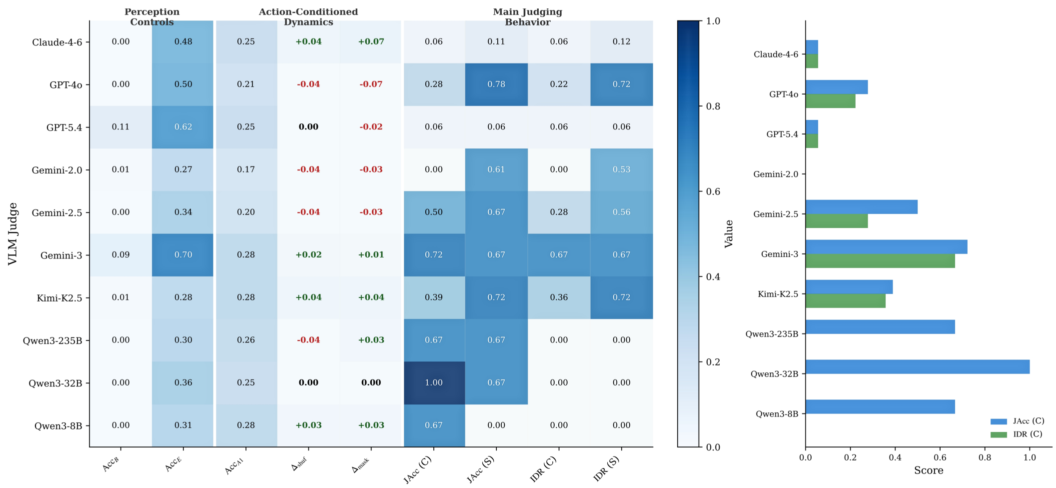
Table 3: **Representation ablation:** Composite vs Split. Mean (Median) Δ = Split$-$Composite. **W**ins/**T**ies/**L**osses count models where Split improves/equals/degrades. The Sig. column reports across-model Wilcoxon signed-rank tests; all comparisons are not significant at $p < 0.05$.

Metric	Composite Mean	Split Mean	Mean Δ	Median Δ	W/T/L	Sig.
$\mathrm{JAcc}_{\mathrm{neu}} \uparrow$	0.433	0.494	+0.061	+0.028	5/2/3	n.s.
$p_{\mathrm{active}} \uparrow$	0.444	0.493	+0.048	+0.031	5/2/3	n.s.
$\mathrm{IDR}_{\mathrm{raw}} \uparrow$	0.163	0.337	+0.173	+0.031	5/5/0	n.s.
$\mathrm{IDR}_{\mathrm{cond}} \uparrow$	0.624	0.720	+0.096	+0.038	4/4/0	n.s.

Key findings: Split representation shows a positive average shift in the main causal-sensitivity metrics in this model suite, with the largest mean change in $\mathrm{IDR}_{\mathrm{raw}}$. However, the across-model Wilcoxon tests are not significant at $p < 0.05$, so we interpret this ablation as evidence that representation can change judging behavior, not as proof that split inputs uniformly improve reliability. For $\mathrm{IDR}_{\mathrm{cond}}$, models with undefined or low-activation conditional IDR are excluded from W/T/L counting according to the reporting rule in Appendix A.

Finding 3: Stability can be degenerate; local controls help explain why. High NS or JCR is not sufficient for reliability. A model can be stable simply because it defaults to a constant verdict or shortcut rule [46]; this is a stability trap when high consistency co-occurs with low $\mathrm{JAcc}_{\mathrm{neu}}$ or low $\mathrm{IDR}_{\mathrm{raw}}$. Figure 5 and Table 4 use Tasks A/B/E to localize possible sources of such failures. Low strict Task B/E scores suggest visual-interface or structured parsing limitations, whereas small drops under Task A action shuffling or masking suggest weak action-conditioned use. Because these controls are strict and local, we use them diagnostically rather than as standalone measures of embodied competence. Full breakdowns and qualitative failure cases are provided in Appendix D.

Summary. Taken together, these results show why SCI requires joint reporting rather than a single leaderboard score. Table 2 shows that accepting successful trajectories, rejecting counterfactual failures, and being active on matched pairs are separable behaviors. Figure 4 and Table 3 further show that changing only the evidence interface can shift verdicts, so improved split-image performance should be interpreted as representation dependence rather than solved reliability. Finally, Figure 5 and Table 4 indicate that some higher-level SCI failures coincide with strict parsing or weak action-conditioning controls, suggesting that causal misgrounding can arise from multiple sources rather than a single uniform failure mode.



(a) Task A/B/E local controls. (b) Failure-source localization.

Figure 5: **Local diagnostic controls and failure localization.** (a) uses Task B/E perception controls and Task A action-conditioned transition diagnostics to test whether models parse local states and use action information. (b) relates these local controls to higher-level SCI failures, helping distinguish interface-level or action-ignorant failures from causal judgment failures.

Table 4: **Local diagnostic controls:** Task B (scene parsing), Task E (object recognition), and Task A (action-conditioned transition). B_Acc = scene parsing accuracy. E_Acc = object recognition accuracy. Acc_A1 = transition accuracy. Δ_{shuf} = shuffle ablation degradation. Δ_{mask} = mask ablation degradation. Strict-interface/parsing limited indicates B_Acc or E_Acc < 0.1 under strict exact-match scoring and should not be interpreted as complete visual blindness. Action-ignorant indicates $\Delta_{\text{shuf}} \approx 0$ under the Task-A ablation.

Model	**B_Acc**	**E_Acc**	**Acc_A1**	Δ_{shuf}	Δ_{mask}	**Failure Hint**
Claude Sonnet 4.6	0.000	0.477	0.252	+0.039	+0.068	P/AI
GPT-4o	0.000	0.495	0.214	-0.039	-0.068	P/AI
GPT-5.4	0.111	0.615	0.252	+0.000	-0.019	AI
Gemini 2.0 Flash	0.007	0.274	0.175	-0.038	-0.025	P/AI
Gemini 2.5 Flash Lite	0.000	0.345	0.204	-0.039	-0.029	P/AI
Gemini 3 Flash	0.090	0.702	0.282	+0.019	+0.010	P/AI
Kimi K2.5	0.007	0.280	0.282	+0.039	+0.039	P/AI
Qwen3-235B	0.000	0.305	0.262	-0.039	+0.029	P/AI
Qwen3-32B	0.000	0.357	0.252	+0.000	+0.000	P/AI
Qwen3-8B	0.000	0.311	0.282	+0.029	+0.029	P/AI

P = strict-interface/parsing limited under exact-match scoring (B_Acc or E_Acc < 0.1), not necessarily complete visual blindness. AI = action-ignorant under the shuffle ablation ($\Delta_{\text{shuf}} \approx 0$).

6 Conclusion

We presented **GridWM-Judge**, a simulator-grounded benchmark for diagnosing VLM judges through the lens of *State-Conditional Invariance*. Instead of asking only whether a judge is accurate or consistent, our framework asks whether it changes its verdict for the right reason: remaining stable when only the presentation changes, but responding when a causal intervention changes the outcome. The MiniGrid diagnostic suite, together with a supplementary MiniWorld external-validity probe, reveals that current VLM judges can be presentation-sensitive, representation-fragile, action-insensitive, or causally misgrounded, and that high stability may reflect a degenerate default strategy rather than reliable judgment. These results position GridWM-Judge as an evaluation-science artifact for auditing VLM judges, clarifying the assumptions under which their verdicts should be trusted, and motivating future judge designs that jointly optimize correctness, presentation invariance, and causal sensitivity.

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Appendix Overview

This appendix provides additional details supporting the main paper. Appendix A gives formal definitions, metric specifications, failure-mode definitions, and reporting conventions. Appendix B describes the MiniGrid environments, data generation pipeline, validator gates, and dataset composition. Appendix C provides task interfaces, prompt templates, and representative examples. Appendix D reports full supplementary results and failure cases. Appendix E presents the 3D MiniWorld extension. Appendix F documents implementation, data release, and reproducibility details. Appendix G discusses scope, limitations, societal impact, and outlook.

A Formal Definitions and Metrics

This appendix expands the State-Conditional Invariance (SCI) framework summarized in Section 3. We define the simulator-grounded trajectory semantics, presentation variables, trajectory variants, and representation formats used by GridWM-Judge; formalize the four diagnostic conditions C1–C4; and provide metric and statistical definitions for the results reported in the main text. We also define the failure modes and degenerate-judge criteria used throughout the analysis, including presentation leakage, representation fragility, action-ignorant heuristics, causal misgrounding, and stability traps. The goal is to make clear what remains fixed under semantics-preserving presentation changes, what changes under causal interventions, and why correctness, consistency, and causal sensitivity must be interpreted jointly.

A.1 State and Presentation Variables

Trajectory groups. Let $g \in \mathcal{G}$ denote a trajectory group generated from the simulator. Each group contains an aligned action sequence, state trajectory, rendered evidence, mission or task text, outcome labels, and audit metadata. We write

$$s_g = (s_{g,0:T}, a_{g,0:T-1}, m_g) \tag{5}$$

for the simulator-grounded trajectory semantics, where $s_{g,0:T}$ is the symbolic simulator-state sequence, $a_{g,0:T-1}$ is the fixed action sequence, and m_g is the mission text when provided. The binary outcome label is computed by the simulator success predicate,

$$\ell_g = h(s_g), \qquad \ell_g \in \{0, 1\}. \tag{6}$$

By convention, $\ell_g = 1$ denotes success and $\ell_g = 0$ denotes failure. The rendered images shown to a model are generated from simulator states but are not treated as privileged state in the model-facing prompt.

Trajectory variants. For each sampled group, GridWM-Judge constructs aligned variants $v \in \{\mathrm{full}, \mathrm{nocue}, \mathrm{cf}\}$ when supported by the corresponding task. The FULL variant is a successful rollout, so $\ell_g^{\mathrm{full}} = 1$. The NOCUE variant removes selected early visual evidence while preserving the success outcome, $\ell_g^{\mathrm{nocue}} = 1$; it is an evidence-removal diagnostic, not a semantics-preserving presentation transform. The CF variant is a counterfactual failure generated by a minimal simulator intervention followed by rerollout under the unchanged action sequence, with $\ell_g^{\mathrm{cf}} = 0$.

Presentations and representations. A VLM judge receives a presentation of trajectory evidence and produces a binary verdict

$$\hat{y}_g^{v,\phi} \in \{0, 1\}, \tag{7}$$

where v denotes the trajectory variant and ϕ denotes the presentation. The presentation variable ϕ can encode prompt framing, temporal display order, visual rendering style, or evidence format. We denote the canonical presentation by

$$\phi_0 = (\text{temp/orig}, \text{vis/clean}, \text{frame/neu}). \tag{8}$$

When discussing representation exchangeability, we write r for the evidence interface, such as a composite storyboard, a split-image sequence, or an oracle state table when available. This separates the C3 comparison from the generic presentation notation without changing the core SCI definition.

State-Conditional Invariance. The core reliability principle is State-Conditional Invariance (SCI): conditioned on the same underlying trajectory semantics s, the judge's verdict should be independent of nuisance presentation variables ϕ,

$$\hat{Y} \perp \Phi \mid S. \tag{9}$$

Equivalently, for a semantics-preserving presentation transform $\phi \in \Phi_{\text{eq}}(s_g)$, a reliable judge should satisfy

$$\hat{y}_g^{\text{full},\phi} = \hat{y}_g^{\text{full},\phi_0}. \tag{10}$$

SCI is therefore not ordinary robustness. It does not require the verdict to be unchanged under all input changes. Instead, it asks whether invariance is appropriate: the verdict should remain stable under presentation changes that preserve s_g, but should change under causal interventions that alter the outcome.

A.2 Four Diagnostic Conditions

We decompose judge reliability into four diagnostic conditions, each instantiated by one or more tasks in GridWM-Judge.

C1: Presentation Invariance. For the same underlying trajectory semantics s_g, the judge should be invariant to nuisance presentation transforms:

$$\forall v \in \mathcal{V}_{\text{fixed}}, \quad \forall \phi \in \Phi_{\text{eq}}(s_g^v), \quad \hat{y}_g^{v,\phi} = \hat{y}_g^{v,\phi_0}. \tag{11}$$

In GridWM-Judge, Φ_{eq} includes temporal presentation transforms (e.g., original versus reversed display order), visual transforms (e.g., clean, noisy, or style-adjusted renderings), and prompt-framing transforms (e.g., neutral, positive, or negative preambles). These transforms are designed to preserve the trajectory semantics and the success label. C1 is mainly tested by Task C through the NS, JCR, Flip, and LES metrics defined below. In the main analysis, C1 is evaluated within a fixed trajectory condition; Full–CF differences are never counted as presentation invariance because they change the underlying outcome.

C2: Action-Conditioned Dynamics. Judge-relevant reasoning should depend on action-conditioned state transitions, rather than on terminal-frame heuristics or superficial visual matching. This condition is conceptually aligned with action-conditioned world-model learning and model-based control [30–33]. Let f denote the implicit transition model needed to predict future states. A minimal requirement is

$$f^{(k)}(s_t, a_{t:t+k-1}) \approx s_{t+k}, \quad k \geq 1. \tag{12}$$

Task A tests the one-step case by asking the model to select the correct next observation o_{t+1} from four candidates given the current observation o_t and action a_t. Action-shuffling and action-masking ablations test whether the model actually uses the action condition.

C3: Representation Exchangeability. The same trajectory semantics may be presented through different but equivalent evidence formats, such as a composite storyboard, a split-image representation, or an oracle state table. This concern is related to evidence-ordering and image-sequence reasoning effects observed in long-context and multimodal settings [24, 25]. A reliable judge should not be bound to a single interface:

$$\hat{y}_g^{v,r,\phi} = \hat{y}_g^{v,r',\phi}, \qquad \text{for the same } (g, v, \phi), \tag{13}$$

whenever these representations encode the same trajectory semantics. In the main experiments, C3 is tested primarily by comparing composite and split-image presentations; oracle state-table variants are optional diagnostics.

C4: Causal Sensitivity. A judge should be insensitive to nuisance transforms but sensitive to causal interventions. For each group, the counterfactual variant changes the simulator state or initial condition so that the same action sequence leads to failure:

$$\begin{aligned} &s_g^{\text{full}} \rightarrow s_g^{\text{cf}}, \\ &\ell_g^{\text{full}} = 1, \qquad \ell_g^{\text{cf}} = 0. \end{aligned} \tag{14}$$

A reliable judge should therefore satisfy

$$\hat{y}_g^{\text{full},\phi_0} = 1 \quad \text{and} \quad \hat{y}_g^{\text{cf},\phi_0} = 0. \tag{15}$$

C4 is mainly tested by Task D, which directly compares full successful trajectories with counterfactual failures.

Task mapping and NoCue. The tasks are not independent leaderboards; they diagnose different components of SCI. Task C tests presentation invariance and representation sensitivity (C1/C3), Task D tests causal sensitivity (C4), Task A tests action-conditioned transition understanding (C2), and Tasks B/E provide perception and state recognition controls. The NoCue variant probes whether the judge depends on early visual cues, but it is not counted as either a semantics-preserving presentation transform for C1 or a causal intervention for C4.

A.3 Metrics Formulation

We define all metrics as empirical estimates over their applicable item sets. Let $\mathbf{1}[\cdot]$ be the indicator function. Unless otherwise stated, $G = |\mathcal{G}|$ denotes the number of trajectory groups in the relevant task, variant set, and representation condition.

Correctness anchors. The main correctness anchor is the neutral full-trajectory judgment accuracy:

$$\mathrm{JAcc}_{\mathrm{neu}} = \frac{1}{G} \sum_{g \in \mathcal{G}} \mathbf{1}\left[\hat{y}_g^{\text{full},\phi_0} = 1\right], \tag{16}$$

where all full trajectories are successful by construction. We also report variant-wise accuracies:

$$\mathrm{Acc}_{\mathrm{full}} = \frac{1}{G} \sum_{g} \mathbf{1}\left[\hat{y}_g^{\text{full},\phi_0} = 1\right], \tag{17}$$

$$\mathrm{Acc}_{\mathrm{nocue}} = \frac{1}{G} \sum_{g} \mathbf{1}\left[\hat{y}_g^{\text{nocue},\phi_0} = 1\right], \tag{18}$$

and

$$\mathrm{Acc}_{\mathrm{cf}} = \frac{1}{G} \sum_{g} \mathbf{1}\left[\hat{y}_g^{\text{cf},\phi_0} = 0\right]. \tag{19}$$

Here $\mathrm{Acc}_{\mathrm{nocue}}$ is diagnostic because the NoCue trajectory preserves the success outcome while removing selected early evidence. We also track the success prediction rate

$$\mathrm{SuccC} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathbf{1}[\hat{y}_i = 1], \tag{20}$$

over the relevant binary judgment set $\mathcal{I}$, as a label-bias indicator.

Presentation invariance: NS and JCR. For a probe family $F \in \{\mathrm{temp}, \mathrm{vis}, \mathrm{frame}\}$, let $\Phi_F(g)$ be the set of non-baseline presentations in that family for group g, and let $\phi_{F,0}$ be the family-specific baseline. The Nuisance Stability score is

$$\mathrm{NS}_F = \frac{1}{|\mathcal{D}_F|} \sum_{(g,\phi) \in \mathcal{D}_F} \mathbf{1}\left[\hat{y}_g^{\text{full},\phi_{F,0}} = \hat{y}_g^{\text{full},\phi}\right], \tag{21}$$

where $\mathcal{D}_F = \{(g,\phi) : g \in \mathcal{G}, \phi \in \Phi_F(g)\}$. The baseline itself is not counted as a trivial self-comparison.

We also compute the Judge Consistency Rate (JCR). Let $\Phi_F^{\text{all}}(g) = \{\phi_{g,F,1}, \dots, \phi_{g,F,M_F}\}$ contain all probe instances in family F, including the family baseline when applicable. Then

$$\text{JCR}_F = \frac{1}{G} \sum_{g \in \mathcal{G}} \mathbf{1}\left[\hat{y}_g^{\text{full},\phi_{g,F,1}} = \dots = \hat{y}_g^{\text{full},\phi_{g,F,M_F}}\right]. \tag{22}$$

The corresponding flip rate is

$$\text{Flip}_F = 1 - \text{JCR}_F. \tag{23}$$

High NS or JCR is not sufficient for reliability, because a constant-verdict judge can be perfectly stable.

Presentation leakage: LES. To measure whether a presentation family systematically shifts verdicts, we estimate a leakage effect size. For each family F, we fit a mixed-effects logistic model over pooled binary predictions:

$$\begin{gathered} \text{logit } \Pr(\hat{Y}_{g,\phi} = 1) = \beta_0 + \sum_{k=1}^{K_F} \beta_{F,k}\, \mathbf{1}[\phi_F = k] + u_g, \\ u_g \sim \mathcal{N}(0, \sigma_g^2). \end{gathered} \tag{24}$$

where u_g is a group-level random intercept and the dummy variables encode the presentation identity within family F. We define

$$\text{LES}_F = \max_k |\beta_{F,k}|. \tag{25}$$

We also report the odds ratio $\exp(\beta_{F,k})$ and the standardized effect size

$$d_{\text{logit},F} = \frac{\text{LES}_F}{\pi/\sqrt{3}}. \tag{26}$$

A large and statistically significant LES_F indicates presentation leakage: the verdict depends systematically on how the same trajectory evidence is presented.

Representation exchangeability. For two equivalent representations r and r', such as composite and split storyboards, we define the Verdict Consistency Coefficient as

$$\text{VCC}_{r,r'} = \frac{1}{|\mathcal{P}_{r,r'}|} \sum_{g \in \mathcal{P}_{r,r'}} \mathbf{1}\left[\hat{y}_g^r = \hat{y}_g^{r'}\right], \tag{27}$$

where $\mathcal{P}_{r,r'}$ is the set of groups evaluated under both representations. When all groups are paired, this reduces to an average over matched groups. In the main tables, this is reported as VCC_{rep} for the composite–split comparison. We can further condition this metric on the baseline representation's predicted verdict:

$$\text{VCC}_{\text{rep}|\text{comp}=1} = \Pr(\hat{y}^{\text{split}} = \hat{y}^{\text{comp}} \mid \hat{y}^{\text{comp}} = 1), \tag{28}$$

and analogously for $\hat{y}^{\text{comp}} = 0$. When oracle state-table representations are available, we compute

$$\text{VCC}_{\text{oracle}} = \Pr(\hat{y}^{\text{oracle}} = \hat{y}^{\text{pixel}}). \tag{29}$$

Low representation consistency indicates representation fragility or a perception-bound judge.

Causal sensitivity: IDR. C4 is measured by whether the judge discriminates successful full trajectories from their counterfactual failures. Define

$$\begin{aligned} A_g &= \mathbf{1}\left[\hat{y}_g^{\text{full},\phi_0} = 1\right], \\ C_g &= \mathbf{1}\left[\hat{y}_g^{\text{cf},\phi_0} = 0\right]. \end{aligned} \tag{30}$$

The raw Intervention Discrimination Rate is

$$\text{IDR}_{\text{raw}} = \frac{1}{G} \sum_{g \in \mathcal{G}} A_g C_g. \tag{31}$$

The number of active groups is

$$G_{\text{active}} = \sum_{g\in\mathcal{G}} A_g, \qquad p_{\text{active}} = \frac{G_{\text{active}}}{G}. \tag{32}$$

Since a model that almost never predicts success on full trajectories cannot meaningfully be tested for full-to-counterfactual discrimination, we also report the conditional IDR:

$$\mathrm{IDR}_{\text{cond}} = \frac{\sum_g A_g C_g}{G_{\text{active}}}, \quad \text{if } G_{\text{active}} > 0. \tag{33}$$

Equivalently,

$$\mathrm{IDR}_{\text{cond}} = \frac{\mathrm{IDR}_{\text{raw}}}{p_{\text{active}}}. \tag{34}$$

We additionally track the reverse error

$$\mathrm{IDR}_{\text{rev}} = \Pr(\hat{y}^{\text{full},\phi_0} = 0 \wedge \hat{y}^{\text{cf},\phi_0} = 1), \tag{35}$$

which should be close to zero for a sensible judge.

Task-specific diagnostic metrics. Task A reports one-step transition accuracy:

$$\mathrm{Acc}_{A1} = \frac{1}{N_A}\sum_{i=1}^{N_A} \mathbf{1}[\hat{c}_i = c_i^\star], \tag{36}$$

where $c_i^\star$ is the correct next-observation choice among four candidates. When multi-step variants are available, we analogously report Acc_{Ak} for k-step prediction. To test whether the model uses the action condition, we compute action-ablation drops:

$$\Delta_{\text{shuffle}} = \mathrm{Acc}_{A1}^{\text{orig}} - \mathrm{Acc}_{A1}^{\text{shuffle}}, \qquad \Delta_{\text{mask}} = \mathrm{Acc}_{A1}^{\text{orig}} - \mathrm{Acc}_{A1}^{\text{mask}}. \tag{37}$$

If these drops are small and statistically insignificant, the model is marked as relying on an action-ignorant heuristic for this diagnostic.

Task B measures structured perception from a single frame. Let $S_B(i) \in [0,1]$ be the component-level score for item i under the canonical schema. We report

$$\mathrm{B\_MeanScore} = \frac{1}{N_B}\sum_{i=1}^{N_B} S_B(i), \qquad \mathrm{B\_Acc} = \frac{1}{N_B}\sum_{i=1}^{N_B} \mathbf{1}[S_B(i) = 1]. \tag{38}$$

Task E analogously reports object/state classification accuracy and mean score:

$$\mathrm{E\_Acc} = \frac{1}{N_E}\sum_{i=1}^{N_E} \mathbf{1}[\hat{z}_i = z_i^\star], \qquad \mathrm{E\_MeanScore} = \frac{1}{N_E}\sum_{i=1}^{N_E} S_E(i). \tag{39}$$

These perception-oriented diagnostics help separate failures caused by visual parsing or interface mismatch from failures in causal judgment.

Metric summary. Table 5 lists the metrics used in the paper and the primary diagnostic question each addresses.

Table 5: **Summary of SCI metrics.** All metrics are empirical estimates over their applicable task-specific item sets.

Metric	Formula sketch	Task(s)	Interpretation
$\mathrm{JAcc}_{\mathrm{neu}}$	$\frac{1}{G}\sum_g \mathbf{1}[\hat{y}_g^{\mathrm{full}}=1]$	D	Neutral full-trajectory correctness anchor
$\mathrm{Acc}_{\mathrm{full}}$	$\frac{1}{G}\sum_g \mathbf{1}[\hat{y}_g^{\mathrm{full}}=1]$	C/D	Variant-wise full success acceptance
$\mathrm{Acc}_{\mathrm{nocue}}$	$\frac{1}{G}\sum_g \mathbf{1}[\hat{y}_g^{\mathrm{nocue}}=1]$	C/D	NoCue success accuracy; evidence-removal diagnostic
$\mathrm{Acc}_{\mathrm{cf}}$	$\frac{1}{G}\sum_g \mathbf{1}[\hat{y}_g^{\mathrm{cf}}=0]$	D; C if CF probes are included	Counterfactual-failure rejection accuracy
SuccC	$\frac{1}{\lvert\mathcal{I}\rvert}\sum_i \mathbf{1}[\hat{y}_i=1]$	C/D	Overall success-verdict rate; label-bias indicator
p_{active}	G_{active}/G	D	Rate at which the judge accepts genuine full successes
$\mathrm{IDR}_{\mathrm{raw}}$	$\frac{1}{G}\sum_g A_g C_g$	D	Joint Full–CF discrimination without conditioning away inactivity
$\mathrm{IDR}_{\mathrm{cond}}$	$\sum_g A_g C_g/G_{\mathrm{active}}$	D	CF rejection conditional on accepting the matched full trajectory
$\mathrm{IDR}_{\mathrm{rev}}$	$\Pr(\hat{y}^{\mathrm{full}}=0 \wedge \hat{y}^{\mathrm{cf}}=1)$	D	Reverse Full–CF ordering error
NS_F	Pairwise agreement with family baseline	C	Stability under a presentation family F
JCR_F	All-presentations-same indicator averaged over groups	C	Per-group consistency across all probes in F
Flip_F	$1-\mathrm{JCR}_F$	C	Presentation-induced verdict flips
LES_F	$\max_k \lvert\beta_{F,k}\rvert$ from logistic model	C	Systematic presentation leakage effect size
$d_{\mathrm{logit},F}$	$\mathrm{LES}_F/(\pi/\sqrt{3})$	C	Standardized logit-scale leakage effect size
$\mathrm{VCC}_{\mathrm{rep}}$	$\frac{1}{\lvert\mathcal{P}\rvert}\sum_g \mathbf{1}[\hat{y}_g^{\mathrm{comp}}=\hat{y}_g^{\mathrm{split}}]$	C; paired D if reported	Composite–split representation exchangeability
$\mathrm{VCC}_{\mathrm{oracle}}$	$\Pr(\hat{y}^{\mathrm{oracle}}=\hat{y}^{\mathrm{pixel}})$	Optional C/D	Agreement between oracle state-table and visual evidence
Acc_{A1}	$\frac{1}{N_A}\sum_i \mathbf{1}[\hat{c}_i=c_i^\star]$	A	One-step action-conditioned transition accuracy
Acc_{Ak}	Analogous k-step choice accuracy	Optional A	Multi-step transition accuracy when available
$\Delta_{\mathrm{shuffle}}, \Delta_{\mathrm{mask}}$	Accuracy drops from original Task A	A	Whether action text affects next-state prediction
B_MeanScore, B_Acc	Mean component score; exact-match accuracy	B	Structured scene-perception control
E_Acc, E_MeanScore	Exact-match accuracy; mean component score	E	Object/state tracking control

A.4 Failure-Mode Taxonomy and Degenerate-Judge Criteria

The metrics above are intended for diagnosis rather than for constructing a single leaderboard. We therefore interpret each metric through a failure-mode taxonomy. A model may exhibit multiple failure modes at once, and degenerate judges are flagged before drawing conclusions from stability or intervention metrics.

- **Presentation leakage** (C1). The verdict changes under semantics-preserving temporal, visual, or framing transforms. This is reflected by low NS_F or JCR_F, high Flip_F, or a non-negligible leakage effect size LES_F. It indicates that the judge is sensitive to how the same trajectory evidence is presented.
- **Representation fragility** (C3). Equivalent evidence formats, such as composite storyboards and split-image sequences, yield inconsistent verdicts. This is reflected by low $\mathrm{VCC}_{\mathrm{rep}}$ or, when available, low $\mathrm{VCC}_{\mathrm{oracle}}$. It indicates that the judge is bound to a particular evidence interface rather than to the underlying trajectory semantics.
- **Action-ignorant heuristic** (C2). Task-A performance does not decrease after action shuffling or action masking. This is reflected by small $\Delta_{\mathrm{shuffle}}$ or Δ_{mask} despite nontrivial Acc_{A1}. It suggests that the model relies on visual matching or terminal-frame shortcuts rather than action-conditioned transition reasoning.
- **Causal misgrounding** (C4). The model fails to distinguish a successful FULL trajectory from its matched counterfactual failure. This is reflected by low $\mathrm{IDR}_{\mathrm{raw}}$ or low/unstable $\mathrm{IDR}_{\mathrm{cond}}$ after checking p_{active} and G_{active}. It indicates that the judge does not reliably respond to an intervention that changes the simulator outcome.
- **Stability trap** (cross-cutting). The model has high consistency under presentation probes but low correctness or low causal sensitivity. In this regime, high NS or JCR reflects a constant verdict or shortcut decision rule rather than reliable judging.

We additionally flag degenerate judges before interpreting SCI metrics:

- **All-fail degenerate**: $p_{\text{active}} < 0.1$. The model rarely accepts genuine successful full trajectories, so high Acc_{cf} or high stability is not evidence of reliability.
- **Success-biased degenerate**: $p_{\text{active}} > 0.9$ and $\Pr(\hat{y}^{\text{cf}} = 1) > 0.9$ (equivalently, $\mathrm{Acc}_{\text{cf}} < 0.1$). The model accepts both full and CF trajectories, so high full-trajectory accuracy does not imply causal sensitivity.
- **Low-activation unstable**: $0 < G_{\text{active}} < 10$. $\mathrm{IDR}_{\text{cond}}$ may be reported descriptively, but it should not be treated as strong evidence because the active set is too small.

These labels are interpretive diagnostics, not additional tasks. They are used to connect the four SCI conditions to the qualitative failure cases reported in Appendix D.

A.5 Statistical Testing and Uncertainty Estimation

Bootstrap confidence intervals. For reported proportions and paired metrics, including $\mathrm{JAcc}_{\text{neu}}$, Acc_{cf}, p_{active}, $\mathrm{IDR}_{\text{raw}}$, $\mathrm{IDR}_{\text{cond}}$, NS, JCR, and VCC, we use percentile bootstrap intervals. Because multiple items may share the same underlying trajectory group, the default procedure is a clustered bootstrap: resample trajectory groups with replacement, include all items associated with the resampled groups, recompute the metric, and report the 2.5th and 97.5th percentiles over the bootstrap estimates. This preserves within-group dependence induced by shared states, actions, and variants.

Paired significance tests. For paired binary comparisons, such as composite versus split predictions on the same group or baseline versus ablated Task-A predictions on the same item, we use McNemar's exact test when the number of discordant pairs is small and the continuity-corrected McNemar test otherwise. If n_{01} and n_{10} are the two discordant counts, the large-sample statistic is

$$\chi^2_{\text{McNemar}} = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}}. \tag{40}$$

For unpaired comparisons across different model runs or disjoint item sets, we use Fisher's exact test for small counts and two-proportion tests for larger counts. Multiple pairwise comparisons are treated as exploratory unless a correction procedure is explicitly stated.

Binomial tests and C4 interpretation. For C4, the primary evidence is the pair $(p_{\text{active}}, \mathrm{IDR}_{\text{raw}})$, because it reveals both whether the model accepts genuine successes and whether it rejects matched CF failures. Conditional testing of $\mathrm{IDR}_{\text{cond}}$ is performed on the active set only: among the G_{active} groups for which $A_g = 1$, the number of successful Full–CF discriminations is $K = \sum_g A_g C_g$. When a pre-specified baseline p_0 is used, we apply an exact one-sided binomial test of $H_0 : p \le p_0$ against $H_1 : p > p_0$. The baseline p_0 should be chosen according to the experimental question, e.g., a chance baseline for forced binary outputs or a conservative zero-discrimination baseline for a diagnostic sanity check.

Low-activation regimes. $\mathrm{IDR}_{\text{cond}}$ can be misleading when the active set is small. We therefore apply the following reporting rule:

1. If $G_{\text{active}} = 0$, $\mathrm{IDR}_{\text{cond}}$ is undefined.
2. If $0 < G_{\text{active}} < 10$, report $\mathrm{IDR}_{\text{cond}}$ only as a descriptive value, mark it as unstable, and do not use it as evidence of causal sensitivity.
3. If $G_{\text{active}} \ge 10$, report $\mathrm{IDR}_{\text{cond}}$ with a clustered bootstrap confidence interval, but still interpret it together with p_{active} and $\mathrm{IDR}_{\text{raw}}$.

In all cases, the main C4 diagnostic plot should show both p_{active} and $\mathrm{IDR}_{\text{raw}}$.

Degenerate-judge criteria. Degenerate-judge criteria are defined in Section A.4. In the statistical workflow below, these criteria are applied before interpreting conditional causal-sensitivity metrics or high stability scores.

Minimal workflow. The statistical workflow used for C4-style diagnostics is:

1. compute A_g, C_g, p_{active}, IDR_{raw}, and IDR_{cond};
2. bootstrap group-level confidence intervals for the reported rates;
3. flag low-activation or degenerate regimes before interpreting IDR_{cond};
4. use paired tests for representation or ablation comparisons;
5. interpret stability metrics only jointly with correctness and causal sensitivity.

B Environment Details and Dataset Generation

This appendix expands Section 4 by documenting the MiniGrid Tier-0 environments, the construction of aligned trajectory variants, the validation gates used before exam generation, and the dataset composition. The purpose of this appendix is to make the simulator-grounded source of truth inspectable: for each sampled seed we retain the symbolic state sequence, action sequence, rendered observations, mission text, outcome labels, and audit metadata, while exposing only the model-facing evidence to the VLM judge. Full JSON/JSONL schemas, repository paths, decoding parameters, and response parsing rules are provided separately in Appendix F.

B.1 MiniGrid Environment Cards

GridWM-Judge uses six deterministic MiniGrid environments as the 2D Tier-0 suite. They cover object interaction, ordered door opening, hazard navigation, memory-dependent choice, and multi-room traversal. Each environment provides a compact symbolic state, an agent-centric rendered observation, a discrete action space, and an unambiguous success predicate. Table 6 summarizes the environment-level causal structure used by the benchmark.

Figure 6 visualizes representative seeds and causal chains for the six MiniGrid environments summarized in Table 6. For each environment, object identities and causal events are inferred from the simulator state rather than from rendered pixels alone. DoorKey records whether the agent is carrying the key and the door state; KeyCorridor resolves the mission target and the color-matched key/door pair; RedBlueDoor checks the red-before-blue ordering; LavaGap identifies the barrier and the unique safe gap; Memory identifies the cue object and the matching terminal object; and MultiRoom tracks

Table 6: **MiniGrid environment cards.** The table lists the causal evidence that the judge should use, the evidence removed in NoCue, and the minimal counterfactual intervention used to create matched failures.

Environment	Causal chain and success condition	NoCue evidence removal	CF intervention
DoorKey	Pick up key $\rightarrow$ unlock/open matching door $\rightarrow$ reach goal. Success requires positive reward and terminal goal entry.	Mask early frames in which the key is visible before pickup.	Relocate or remove the goal at $t = 0$ so that the unchanged action sequence no longer reaches the true goal.
KeyCorridor	Pick up color-matched key $\rightarrow$ return to locked door $\rightarrow$ unlock door $\rightarrow$ reach the mission target object.	Mask the door-matched key in the pre-pickup window.	Remove the mission target object before rollout, so the same action sequence cannot satisfy the pickup objective.
RedBlueDoor	Open red door $\rightarrow$ open blue door $\rightarrow$ pass through the ordered-door sequence. Opening blue before red is invalid.	Mask the red door before the first red-door interaction; blue-door evidence is retained.	Replace or block the blue-door passage before rollout, preserving the early red-door subtask but preventing final success.
LavaGap	Identify the single safe gap in the lava barrier $\rightarrow$ cross through the gap $\rightarrow$ reach goal without stepping on lava.	Mask visible lava evidence in early pre-crossing frames.	Relocate or remove the goal outside the planned path while keeping the safe-crossing behavior physically valid.
Memory	Observe cue object $\rightarrow$ traverse corridor $\rightarrow$ choose the matching terminal object.	Mask the cue object before corridor entry.	Swap the correct and distractor terminal objects before rollout, so the same remembered choice becomes wrong.
MultiRoom	Traverse a sequence of rooms, opening intervening doors, then enter the goal room and reach the goal.	Mask early door/goal evidence before entering the goal-containing room.	Remove the final goal object before rollout, preserving room traversal but preventing terminal success.

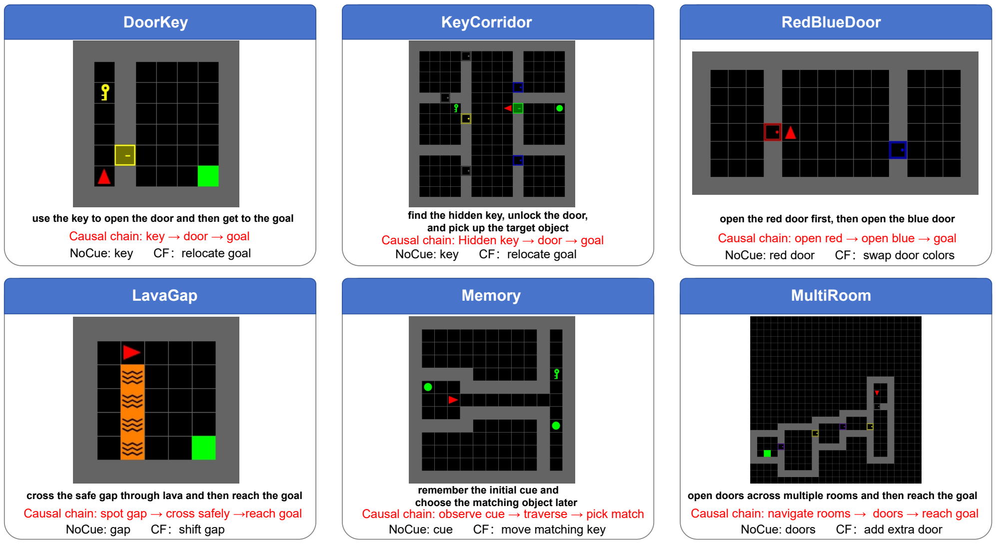


Figure 6: **Representative MiniGrid environment cards.** Each panel shows one canonical seed, the mission text, and a short causal-chain annotation.

door toggles, room transitions, and entry into the goal room. These symbolic event traces are used only for generation and validation, not as model-facing inputs.

B.2 Dataset Generation Pipeline

The atomic unit for Tasks C and D is a trajectory group $g = (s_{0:T}, a_{0:T-1}, m, \ell, \eta)$, containing a simulator state sequence, action sequence, mission text, outcome label, and audit metadata. A trajectory group contains a successful Full rollout, an evidence-removal NoCue variant, and a counterfactual CF variant. Tasks A, B, and E are derived from the same validated simulator rollouts but expose more localized transition, perception, or prefix-checkpoint interfaces, as described in Appendix C. Figure 7 shows the full generation pipeline from environment seed to validated model-facing exam items.

Stage 1: successful full rollout. For each environment seed, we reset the simulator with a deterministic layout and use an environment-specific shortest-path planner to construct a successful rollout. The planner operates over symbolic states and inserts required interaction actions, such as pickup and toggle. Rollouts with truncation, missing critical events, or ambiguous terminal outcomes are rejected before any exam item is produced.

Stage 2: evidence-removal variant. The NoCue variant keeps the same action sequence and the same simulator outcome as Full, but masks selected early rendered tiles that reveal the causal evidence. The masked frames must occur strictly before the first interaction with the target evidence. Thus NoCue is an outcome-preserving evidence-removal diagnostic: it tests whether the judge relies on early causal evidence, but it is not counted as a semantics-preserving presentation transform for C1 or as a causal intervention for C4. Table 7 lists the environment-specific evidence targets, mask budgets, and alignment thresholds used for NoCue construction. Figure 8 illustrates a representative NoCue audit, showing how the target evidence is masked while preserving the rollout outcome.

Stage 3: counterfactual failure variant. The CF variant applies a minimal simulator intervention before action execution and then rerolls the unchanged action sequence. Depending on the environment, the intervention removes, relocates, swaps, or blocks the causal target. The validation contract requires the counterfactual rollout to be a physically valid execution of the same action sequence and to fail for the intended causal reason. Exact positional identity with Full is required for NoCue; for CF, state divergence is allowed only when it is induced by the declared intervention under the unchanged actions.

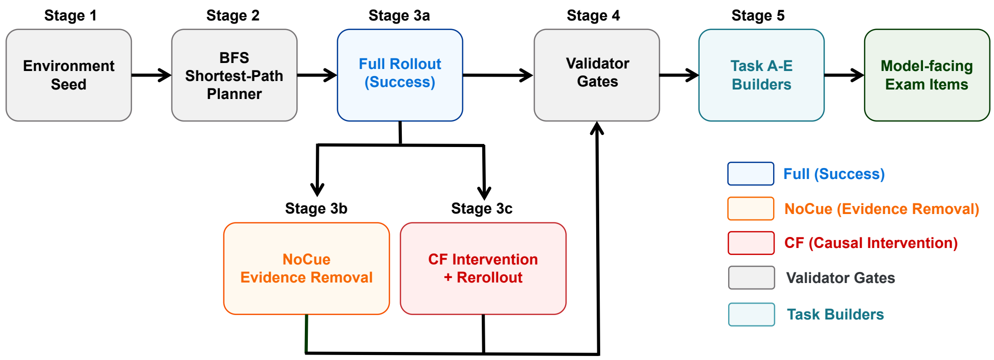


Figure 7: **Dataset generation pipeline from environment seed to validated exam item.** Each seed first produces a successful Full rollout; the aligned NoCue and CF variants are then constructed, validated, and converted into task-specific exam items. The model-facing prompts contain only rendered evidence and task instructions, while symbolic states, labels, and intervention metadata remain audit-only.

Table 7: **Per-environment NoCue targets and masking constraints.** The mask budget is intentionally conservative so that masking removes the target evidence without changing the overall trajectory presentation.

Environment	Masked target evidence	Max masked frames	Alignment threshold
DoorKey	Key before pickup	5	0.70
KeyCorridor	Color-matched key before pickup	5	0.70
RedBlueDoor	Red door before first red-door toggle	5	0.70
LavaGap	Lava evidence before safe crossing	4	0.70
Memory	Cue object before corridor entry	10	0.70
MultiRoom	Door/goal evidence before goal-room entry	12	0.70

Stage 4: rendering and exam assembly. Validated rollouts are rendered into the presentation formats used by the task interfaces. Task C uses event-anchored sparse storyboards and presentation probes such as temporal order and visual style changes. Task D uses matched Full/CF trajectory evidence for causal-sensitivity tests. Task E uses prefix-only checkpoint evidence and deliberately hides terminal frames. The detailed prompt templates, split-image/composite layouts, and example questions are provided in Appendix C.

B.3 Validator Gates and Quality Control

All generated groups pass validation before they enter the benchmark. The validators prevent data-generation artifacts from being mistaken for VLM-judge failures. Table 8 summarizes the gates used in the release pipeline.

Environment-specific semantic checks instantiate E1. DoorKey requires key pickup before door opening; KeyCorridor requires the color-matched key before the locked-door interaction; RedBlueDoor requires the red door to be opened before the blue door; LavaGap requires crossing through the safe gap and never stepping on lava; Memory requires cue observation before the terminal choice; and MultiRoom requires the expected sequence of door toggles and goal-room entry.

B.4 Dataset Composition

We report the benchmark in terms of exam rows rather than only raw trajectories, because the same validated trajectory group can instantiate multiple variants and presentation conditions. Table 9 summarizes the paper-facing MiniGrid benchmark used in the main experiments. The table describes the diagnostic role of each task, the source environments, the number of model-facing exam rows, and the variant or probe structure used to construct those rows.

The five tasks serve complementary diagnostic roles rather than forming a single undifferentiated leaderboard. Task A probes local action-conditioned transition understanding; Task B and Task E

Table 8: **Validator gates.** The gates jointly enforce action alignment, outcome correctness, physically valid rerollout, and correct isolation of model-facing evidence from audit-only state.

Gate	Purpose	Rule	Scope
D1	Action identity	The action IDs are identical across matched variants in the same group.	all variants
D2	Length consistency	Frames and states align with the action sequence: $\lvert s_{0:T}\rvert = \lvert a_{0:T-1}\rvert + 1$.	all variants
D3	Input isolation	Model-facing inputs contain only rendered evidence, mission/action text, and task instructions; labels, rewards, states, and intervention metadata remain audit-only.	all tasks
D4	Outcome contract	Full succeeds, NoCue preserves the Full outcome, and CF fails without truncation artifacts.	C/D
D5	No interaction masking	NoCue never masks the frame in which the agent is facing or interacting with the target evidence.	NoCue
D6	Physics validity	NoCue preserves the exact state trajectory; CF is a valid rerollout under the same action sequence, with divergence allowed only as a consequence of the declared intervention.	all variants
E1	Causal-chain completeness	Environment-specific critical events occur in the required order, e.g., key before door, red before blue, gap crossing without lava contact, cue before terminal choice.	all envs
E2	State encoding validity	Symbolic states use valid MiniGrid object, color, and state indices and remain aligned with rendered frames.	all variants
N1	Mask coverage/alignment	Masked pixels overlap the intended target tile with mean IoU at least 0.70 and satisfy the global mask budget.	NoCue
CF1	Intervention audit	The counterfactual target is removed, relocated, swapped, or blocked according to the environment card, and the failure is attributable to that intervention.	CF

Figure 8: **Example NoCue audit.** The panels show the original frame, the target-tile mask overlay, and the resulting NoCue masked frame used to remove selected early evidence while preserving the rollout outcome.

provide perception and state recognition controls; Task C instantiates presentation and representation probes; and Task D provides the matched intervention setting for causal-sensitivity analysis. For Task D, the primary C4 comparison is the paired Full/CF judgment, while retained NoCue rows are used only as auxiliary evidence-removal diagnostics. Model-specific run information, aggregate exclusions, and per-model results are reported separately in the experimental appendices.

B.5 Data Format and Access Boundary

Each exam item separates the evidence shown to the VLM judge from the simulator metadata used for auditing and scoring. The model-facing payload consists of the rendered image or image sequence, the mission or action text when required by the task, the task instruction, and the requested answer format. Audit-only fields include the ground-truth label, success flag, reward, termination status, symbolic state trace, action IDs, random seed, NoCue metadata, and CF intervention metadata. This separation is a core part of the benchmark design: the VLM must judge from the same visual/textual evidence that appears in the prompt, while the simulator state remains available for reproducible validation.

Table 9: **Dataset composition for the paper-facing MiniGrid benchmark.** Counts are reported at the exam-row level, since a single validated trajectory group may produce multiple model-facing items through variants, presentation probes, or task-specific interfaces.

Task	Diagnostic role	Environments	Exam rows	Variant/probe structure
A	Action-conditioned next-observation prediction	DoorKey	103	baseline / action-shuffle / action-mask
B	Single-frame perception and interface control	selected validated MiniGrid frames	144	structured scene/object parsing
C	Presentation invariance and representation sensitivity	KeyCorridor, MultiRoom, RedBlueDoor	324	3 envs $\times$ 6 groups/env $\times$ 3 variants $\times$ 2 temporal probes $\times$ 3 visual probes
D	Causal sensitivity under matched intervention	all six environments	108	6 envs $\times$ 6 groups/env $\times$ 3 variants; FULL/CF form the primary C4 pair and NOCUE is auxiliary
E	Prefix/checkpoint perception control	KeyCorridor and DoorKey retained-control items	48	prefix-only checkpoints; identity/status queries

A minimal exam item therefore has the following logical structure:

```
{
  "uid": "C.keycorridor.s000001.full.orig.clean",
  "task": "C",
  "env_task": "keycorridor",
  "group_id": "s000001",
  "variant": "full",
  "presentation": {
    "temporal": "orig",
    "visual": "clean",
    "rep": "composite"
  },
  "model_input": {
    "image": "...",
    "prompt": "..."
  },
  "audit_only": {
    "label": 1,
    "success": true,
    "state_seq": "...",
    "cf_meta": null
  }
}
```

The exact field names, file locations, dataset card, license, model identifiers, API parameters, and response parsing rules are documented in Appendix F. Keeping those implementation details in Appendix F avoids duplicating schema definitions across appendices and keeps Appendix B focused on the data-generation logic that supports the main SCI claims.

C Task Interfaces and Prompt Templates

GridWM-Judge contains five diagnostic task interfaces. This appendix specifies what each model-facing item shows to the VLM judge, what answer format is expected, and how the gold answer is obtained from simulator-grounded metadata. Appendix A gives the mathematical definitions of the metrics, Appendix B describes trajectory generation and validation, and Appendix F gives the full machine-readable schemas, decoding settings, and unified output parsing rules. Throughout this appendix, *model-facing* fields denote information provided to the VLM judge, whereas labels, rewards, symbolic states, intervention records, and audit metadata are used only for scoring and validation. Table 10 summarizes the model-facing evidence, required answer format, and diagnostic role for the five task interfaces. Figure 9 gives representative model-facing layouts for the five tasks, illustrating the difference between local controls, presentation probes, transition prediction, and causal trajectory judgment.

Table 10: **Summary of the five task interfaces.** Appendix C describes the model-facing input/output contract; Appendix A defines the metrics computed from these outputs.

Task	Model-facing evidence	Required answer	Diagnostic role
B	Single MiniGrid frame with a generic scene-description instruction	Structured JSON scene description or component labels	Perception and interface control
E	Prefix-only sparse storyboard before the terminal state	JSON label for the first picked object or checkpoint status	Object/state tracking control
C	Sparse trajectory storyboard with presentation, framing, and representation probes	Binary success/failure verdict	Presentation invariance; representation exchangeability
A	Current observation, action condition, and four candidate next observations	Multiple-choice next-state label	Action-conditioned transition diagnostic
D	Full-trajectory evidence for matched FULL and CF variants	Binary success/failure verdict	Causal sensitivity under intervention

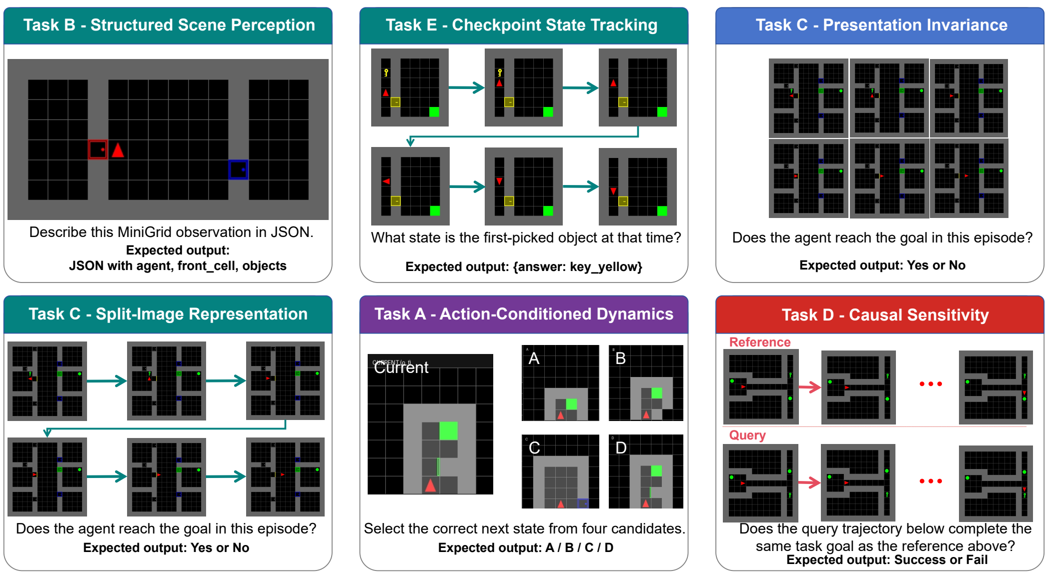


Figure 9: **Representative model-facing interfaces for the five GridWM-Judge tasks.** Task B uses a single-frame structured perception query; Task E uses a prefix-only checkpoint storyboard; Task C presents the same trajectory through composite or split-image evidence formats; Task A asks for the correct action-conditioned next observation; and Task D compares matched Full/CF trajectory judgments. Audit-only simulator states, labels, rewards, and intervention metadata are not shown to the model.

C.1 Task B and E: Perception Controls

Tasks B and E are local controls. They are not intended to establish SCI by themselves; instead, they test whether the model can parse the visual interface before higher-level failures in Tasks C and D are interpreted as presentation or causal-reasoning failures.

Task B: structured scene perception. Task B presents a single egocentric MiniGrid observation and asks the model to produce a structured description of the local state. The item includes a generic instruction explaining the requested fields, but it does not include the mission, action sequence, reward, terminal flag, or symbolic state.

> **Prompt template.** You are given one MiniGrid observation. Describe the visible local state using JSON only. Include the agent direction, whether the agent is carrying an object, the front cell, and visible objects. Do not infer hidden objects.

The canonical output is a JSON object with fields such as `agent`, `front_cell`, and `objects`. A representative schema-level example is:

```
{
  "agent": {
    "dir": "right",
    "carrying": null
  },
  "front_cell": {
    "type": "key",
    "color": "yellow",
    "state": null
  },
  "objects": [
    {"type": "key", "color": "yellow", "relative_pos": "front"},
    {"type": "door", "color": "yellow", "state": "locked"}
  ]
}
```

Gold answers are derived from the simulator state associated with the rendered frame. We report both an exact-match accuracy and a component-level score when available, so that a partially correct parse is not conflated with a completely invalid response. The precise component mapping and JSON parser are documented in Appendix F.

Task E: checkpoint-style long-range state tracking. Task E presents a prefix-only sparse storyboard from frame 0 through a checkpoint t_c, where t_c is strictly before the terminal frame. The model must answer an object-identity or object-status question without using a terminal shortcut. The item does not expose symbolic state, reward, success labels, or future frames after t_c.

> **Identity-query template.** Below is a MiniGrid trajectory storyboard showing the trajectory up to the last shown frame only. Earlier in the shown trajectory, what was the first object the agent picked up? Answer using JSON only: `{"answer":"<label>"}`.

> **Status-query template.** Below is a MiniGrid trajectory storyboard showing the trajectory up to the last shown frame only. By the last shown frame, what is the status of the first object the agent picked up earlier? Answer using JSON only: `{"answer":"<label>"}`.

The identity label space contains object–color labels such as `key_yellow`, `key_green`, `key_purple`, `ball_yellow`, `ball_green`, and related colors appearing in the canonical artifact. The status label space contains `holding`, `dropped`, and `reacquired_and_holding`. Gold labels are derived from the simulator's object-tracking events before or at t_c: the first pickup event determines the tracked object, and the object's checkpoint inventory/grid status determines the status label.

C.2 Task C: Presentation Invariance Tests

Task C is the main interface for presentation-aware binary judging. It asks whether a trajectory satisfies the task goal, while varying how the same trajectory evidence is presented. In the SCI terminology, temporal order, visual style, and prompt framing instantiate C1 presentation probes, while composite versus split-image evidence formats instantiate the C3 representation exchangeability probe. These changes should not alter the verdict when the underlying simulator semantics are unchanged.

Input and answer contract. The model sees a sparse storyboard built from event-anchored keyframes and a short success/failure question. The mission text is included when the item uses an environment-level goal. The action sequence, symbolic states, terminal flag, reward, intervention metadata, and gold label are not shown.

> **Neutral prompt template.** Below is a storyboard from one MiniGrid trajectory.
> Task goal: `{mission_string}`
> Did the trajectory satisfy this goal? Answer with ONLY: Yes or No.

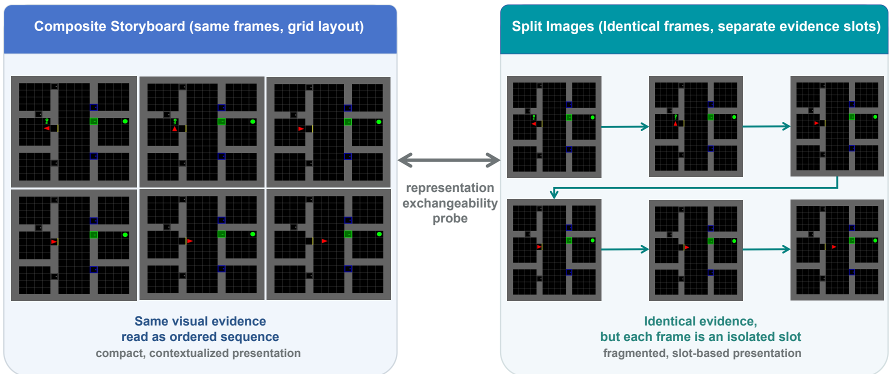


Figure 10: **Composite storyboard versus split-image representation for Task C.** The left panel arranges selected event-anchored keyframes into a compact grid (composite format). The right panel presents the same keyframes as individually numbered images (split-image format). Both formats convey identical trajectory evidence; only the evidence layout differs. The model receives the same prompt and mission text for both formats. A reliable judge should give the same verdict regardless of which representation is used.

Table 11: **Task C probe axes and their interpretation.** Presentation and representation probes are evaluated within a fixed trajectory condition. When CF rows are present, Full–CF differences are treated as causal/outcome changes rather than semantics-preserving presentation changes.

Axis	Values	Interpretation
Trajectory condition	Full, NoCue, CF	Changes the evidence condition or the underlying outcome; not all levels are semantics-preserving.
Temporal order	original, reversed	C1 presentation probe over frame ordering.
Visual style	clean, noisy, style-transformed	C1 presentation probe over low-level rendering perturbations.
Prompt framing	neutral, positive, negative	C1 presentation probe over wording or prior-framing changes, without changing the task goal or evidence.
Representation	composite storyboard, split images	C3 equivalent-evidence-format probe for representation exchangeability.

The output is mapped to a binary verdict, with `Yes`/`Success` corresponding to success and `No`/`Fail` corresponding to failure. The exact parser and invalid-response handling are specified in Appendix F.

Probe axes. Table 11 summarizes the Task C trajectory, presentation, and representation axes. Task C items are organized by trajectory condition and presentation condition. The trajectory condition may be Full, NoCue, or CF; only the presentation and representation axes are used as semantics-preserving invariance probes within a fixed trajectory condition. NoCue removes early visual evidence while preserving the successful outcome, and CF changes the underlying outcome through a simulator intervention. Therefore, full–counterfactual changes should be interpreted as causal sensitivity rather than presentation invariance.

Storyboard and representation formats. Figure 10 illustrates the composite storyboard and split-image representations used for the C3 exchangeability probe. For composite inputs, the selected keyframes are arranged into a compact grid and read left-to-right, top-to-bottom unless the temporal probe reverses the order. For split-image inputs, the same selected frames are supplied as separate images with the same prompt and frame ordering. The selected frames are event-complete: they prioritize causally relevant events such as first object visibility, pickup, door toggling, hazard crossing, goal visibility, and terminal evidence. Matched conditions for the same trajectory group should use the same frame indices whenever the corresponding frames exist, so that differences in verdicts can be attributed to the intended presentation or causal variable rather than accidental frame selection.

Visual and framing variants. The `clean` condition uses the original rendering. The `noisy` condition applies mild deterministic image noise that should not alter the MiniGrid state. The `style` condition applies deterministic color, contrast, brightness, or sharpness adjustments. Prompt-framing variants should change only the wording around the same evidence and goal; they must not add hints, labels, rewards, or causal explanations. For example, positive framing can say that the trajectory may have succeeded, whereas negative framing can say that it may have failed, but both must ask the same binary question over the same visual evidence.

Framing prompt templates. All framing variants use the same images, mission text, and answer format; only the preamble changes. The neutral framing is:

> Below is a storyboard from one MiniGrid trajectory.
> Task goal: `{mission_string}`
> Did the trajectory satisfy this goal? Answer with ONLY: Yes or No.

The positive framing is:

> The trajectory may show successful progress toward the task goal. Please judge only from the visual evidence.
> Task goal: `{mission_string}`
> Did the trajectory satisfy this goal? Answer with ONLY: Yes or No.

The negative framing is:

> The trajectory may contain a failure to complete the task goal. Please judge only from the visual evidence.
> Task goal: `{mission_string}`
> Did the trajectory satisfy this goal? Answer with ONLY: Yes or No.

C.3 Task A: Action-Conditioned Dynamics

Task A tests whether the VLM judge uses the action condition to identify the correct next observation. Unlike Tasks C and D, Task A is a local transition interface rather than a trajectory-level success/failure judgment.

Input and prompt. The item shows a current observation o_t, an action condition, and four candidate next observations labeled A–D. The candidates are rendered from MiniGrid states and are arranged as a multiple-choice panel. One candidate is the true next observation o_{t+1} under the action; the other candidates are visually similar negatives. Figure 11 shows the multiple-choice layout used for Task A; the correct/wrong annotations are included only for illustration and are never shown to the model.

> **Prompt template.** Action: `{action_string}`.
> On the left is the current state. On the right are four candidate next states (A, B, C, D). Select the correct next state after executing the action. Answer with ONLY one letter: A, B, C, or D.

Candidate construction. Negative candidates are selected to avoid trivial recognition. Same-trajectory negatives are preferred, because they share the same environment, objects, and visual style but correspond to different timesteps. If needed, cross-trajectory negatives with similar local configurations are used as fallback. Candidate order is randomized with the gold candidate uniformly placed among A–D.

Action ablations. Task A supports three action conditions. In the baseline condition, the true action is shown. In the shuffle condition, the action field is replaced by a mismatched action sampled from the action label space. In the mask condition, the action field is replaced with an unknown-action token. The ablated interfaces should not expose an action identifier that trivially recovers the original action. Large accuracy drops from baseline to shuffle/mask indicate that the model relies on action-conditioned dynamics; small drops suggest an action-ignorant visual heuristic.

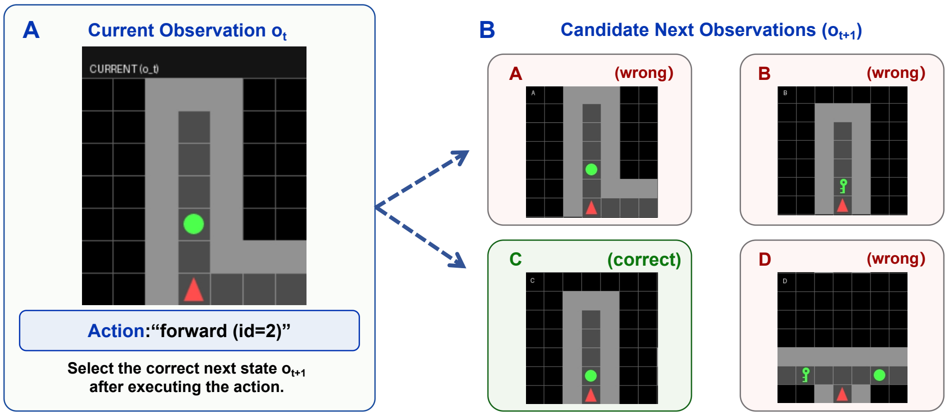


Figure 11: **Layout of a Task A multiple-choice item.** The current observation o_t and the action text are shown on the left. Four candidate next observations (A–D) are rendered on the right. One candidate is the true next observation o_{t+1}; the others are visually similar hard negatives. Dashed annotations indicate which candidate is the correct transition and which negatives share the same trajectory or similar visual configurations. The gold answer is the letter of the correct candidate. Correct/wrong annotations are shown only for illustration and are not included in model-facing inputs.

Ablated action templates. The baseline condition displays the true action as `Action: {action_string}`. In the shuffle condition, the same field is replaced by a different action sampled from the MiniGrid action set: `Action: {shuffled_action_string}`. In the mask condition, the action field is replaced by `Action: unknown`. The image candidates and answer format remain unchanged across these conditions.

C.4 Task D: Causal Sensitivity Tests

Task D is the primary interface for causal sensitivity. It compares matched successful Full trajectories against CF trajectories generated by a minimal simulator intervention and rerollout under the unchanged action sequence. The intended question is whether the judge changes its verdict when the underlying task outcome changes.

Input and prompt. The canonical Task D item presents a full temporal montage rather than a sparse storyboard. Frames are arranged in temporal order and annotated by timestep. The item may include the environment mission or task goal when this is part of the model-facing exam item; it never exposes symbolic states, rewards, success labels, intervention metadata, or validation logs.

> **Prompt template.** Below is a time-ordered MiniGrid trajectory storyboard, arranged from left to right and top to bottom. Judge whether the agent ultimately completed the task. Answer with ONLY: Success or Fail.

Gold answer and scoring. The gold answer is `Success` for the matched successful Full trajectory and `Fail` for the corresponding CF trajectory. The main C4 metrics are computed over paired Full–CF groups and are defined in Appendix A. If a NoCue version is included in the Task D artifact, it should be treated as an evidence-removal diagnostic rather than as the causal-intervention pair used for C4.

Representative paired case. Figure 12 illustrates a representative DoorKey Full/CF pair. The pair keeps the same action sequence. In the Full trajectory, the agent picks up the key, unlocks the door, and reaches the goal. In the matched CF trajectory, the initial intervention relocates or removes the goal so that the same action sequence no longer satisfies the task. A reliable judge should accept the Full montage and reject the matched CF montage, while not changing its verdict merely because the same montage is shown in a different but semantically equivalent representation.

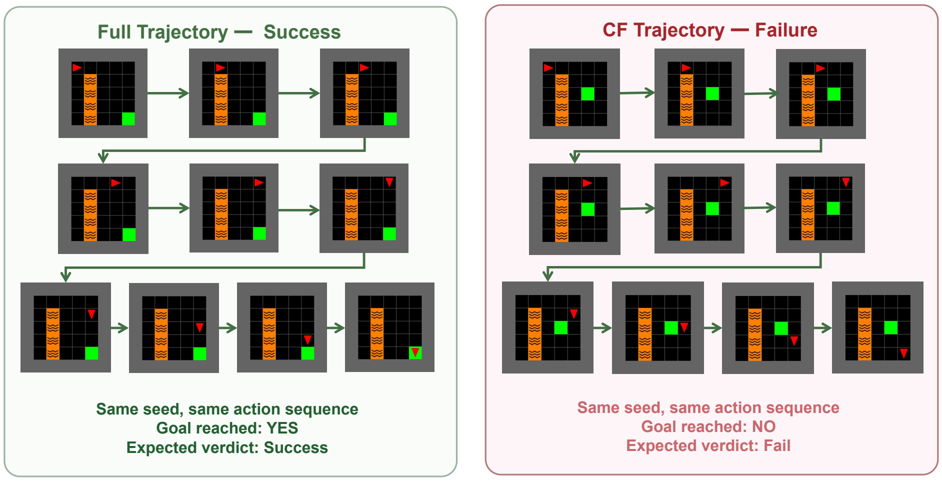


Figure 12: **Illustration of a matched Full/CF pair for Task D (DoorKey example).** Both panels show the same action sequence in the same visual format. In the Full trajectory (left), the goal is reachable and the agent completes the task; the gold verdict is `Success`. In the matched CF trajectory (right), a minimal intervention relocates the goal before the episode starts; the same action sequence now fails. The gold verdict is `Fail`. A reliable judge should accept the Full montage and reject the CF montage.

D Supplementary Results and Analysis

This appendix reports supplementary results for the 2D MiniGrid benchmark. The main text focuses on compact SCI-facing summaries; here we provide additional per-model metrics, representation-consistency results, environment-level breakdowns, qualitative failure cases, and exploratory LLaVA-family runs. The goal is to support the main claims without turning Appendix D into a full dump of all machine-readable result columns.

D.1 Compact Per-Model MiniGrid Metrics

For the ten evaluated VLM judges in the main model suite, we computed a suite of supplementary diagnostic metrics on the 2D MiniGrid benchmark. The main text reports the compact SCI-facing summary, while this appendix provides additional per-model metrics and breakdowns. Table 12 gives a compact overview averaged over composite and split representations; subsequent tables and figures report representation-specific, environment-specific, and qualitative analyses.

We report only the most interpretable metrics in the paper. Additional columns from the final result CSV, such as visual-probe stability, flip rates, per-frame effect sizes, and parsing diagnostics, are provided in the machine-readable supplement.

Several patterns emerge from Table 12. First, no single model dominates all diagnostic axes: for instance, GEMINI 3 FLASH has the strongest Task-D-style counterfactual-failure accuracy, whereas KIMI K2.5 has the strongest average stability metrics. Second, high success acceptance or high counterfactual accuracy alone is insufficient for SCI reliability; these values must be interpreted together with p_{active}, IDR_{raw}, and IDR_{cond}. Third, the aggregate numbers remain far from saturated, with mean JAcc_{neu} around 0.29 and mean IDR_{cond} around 0.24.

D.2 Representation Consistency

The main paper reports that composite and split-image evidence formats can change judge behavior even when the underlying trajectory semantics are fixed. Table 13 provides a compact representation-consistency summary using VCC. Higher values indicate that the judge gives the same verdict across equivalent evidence formats; 1.0 would indicate perfect representation exchangeability.

Table 12: **Compact per-model overview of supplementary 2D MiniGrid metrics.** Values are averaged over composite and split representations to provide a single summary row per model. This table is intended as an overview rather than as a replacement for representation-specific analysis. See Appendix A for metric definitions.

Model	JAcc_neu	Acc_full	Acc_cf	SuccC	SuccD	IDR_raw	IDR_cond	NS	JCR	LES
Claude Sonnet 4.6	0.25	0.86	0.69	0.94	0.69	0.38	0.25	0.87	0.75	0.28
Gemini 2.0 Flash	0.36	0.95	0.20	0.88	0.20	0.54	0.08	0.65	0.55	0.20
Gemini 2.5 Flash Lite	0.24	0.94	0.50	0.94	0.50	0.39	0.25	0.87	0.75	0.24
Gemini 3 Flash	0.28	0.92	0.78	0.95	0.78	0.40	0.26	0.88	0.77	0.30
GPT-4o	0.35	0.94	0.50	0.94	0.50	0.47	0.27	0.88	0.75	0.29
GPT-5.4	0.26	0.78	0.66	0.78	0.66	0.30	0.24	0.85	0.75	0.23
Kimi K2.5	0.36	0.94	0.54	0.94	0.54	0.48	0.31	0.89	0.78	0.37
Qwen3-8B	0.26	0.90	0.50	0.90	0.50	0.37	0.25	0.87	0.75	0.27
Qwen3-32B	0.25	0.90	0.50	0.90	0.50	0.35	0.24	0.87	0.75	0.27
Qwen3-235B	0.24	0.94	0.50	0.94	0.50	0.36	0.25	0.88	0.75	0.29
Mean	0.29	0.91	0.54	0.90	0.54	0.40	0.24	0.87	0.76	0.28

Table 13: **Representation-consistency metrics.** VCC measures the fraction of matched representation pairs for which the judge gives the same verdict. We report VCC for Task C pooled across Full and CF variants, and the analogous representation-consistency score for Task D. NoCue is treated as an evidence-removal diagnostic rather than a semantics-preserving representation change.

Model	$VCC^{\text{pooled}}_{\text{rep},C}$	$VCC_{\text{rep},D}$
Claude Sonnet 4.6	0.69	0.71
Gemini 2.0 Flash	0.66	0.65
Gemini 2.5 Flash Lite	0.66	0.66
Gemini 3 Flash	0.73	0.74
GPT-4o	0.71	0.72
GPT-5.4	0.66	0.70
Kimi K2.5	0.76	0.78
Qwen3-8B	0.66	0.68
Qwen3-32B	0.66	0.68
Qwen3-235B	0.66	0.70
Mean	0.68	0.70

The mean VCC values are around 0.68–0.70, indicating that representation consistency is imperfect across the model suite. These results support the need to report representation exchangeability separately from both plain accuracy and causal-sensitivity metrics.

D.3 Environment-Level Breakdown

The aggregate results above may hide environment-specific failure patterns. Figure 13 therefore reports an environment-level breakdown for the Task C MiniGrid settings. We restrict this plot to the three environments that share the same Task C presentation and representation probe structure—KeyCorridor, MultiRoom, and RedBlueDoor—so that the environment comparison is not confounded by task-specific coverage differences. Scores are averaged over composite and split representations.

D.4 Representative Failure Cases

To complement the aggregate metrics, Figure 14 shows representative qualitative failures selected from the final frozen MiniGrid manifest. The purpose is not to introduce new metrics, but to make the failure-mode taxonomy in Appendix A concrete.

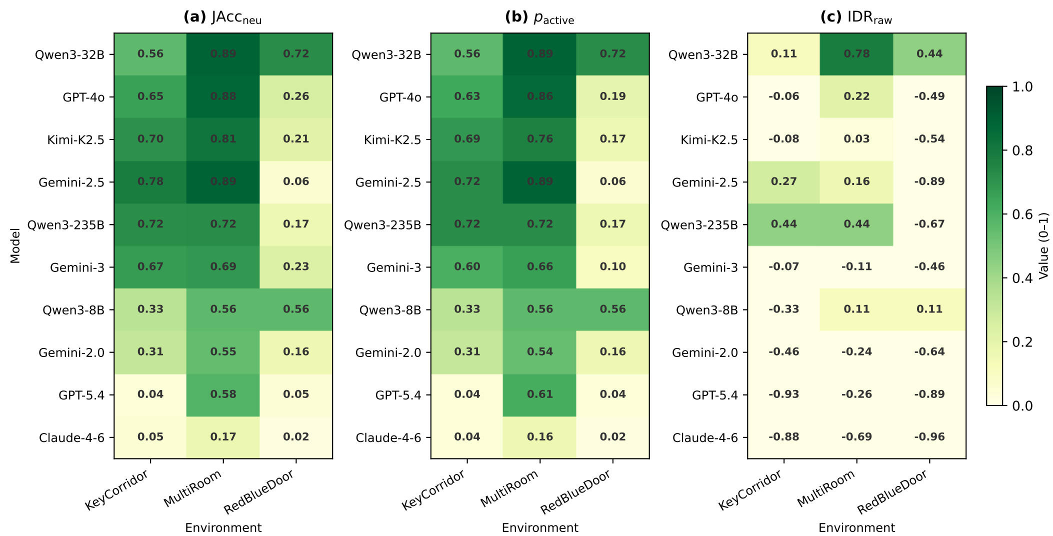


Figure 13: **Environment-level MiniGrid breakdown for Task C.** The figure reports $\mathrm{JAcc}_{\mathrm{neu}}$, p_{active}, and IDR_{raw} for KeyCorridor, MultiRoom, and RedBlueDoor, averaged over composite and split representations. We focus on these three environments because they share the same Task C presentation and representation probe structure. The breakdown shows that model reliability varies across environment types, rather than being explained by a single aggregate score.

D.5 Exploratory LLaVA-Family Results

For completeness, we also evaluated LLaVA-family multimodal models under the same MiniGrid diagnostic protocol. These runs are exploratory and are not used for the main claims, because they extend the model suite beyond the primary judge set discussed in the main text. Table 14 summarizes the compact results.

The exploratory LLaVA results are substantially below the main judge suite on both correctness and intervention-discrimination metrics. The main failure surface appears to combine visual-interface parsing difficulty with weak causal-discrimination ability. We therefore report these results only as a reference point for open VLM-family models, not as evidence for the primary SCI claims.

D.6 Discussion and Recommendations

The supplementary experiments reported in this appendix support the main paper's conclusions. In particular, they show that: (i) state-conditional invariance remains a challenging requirement even under controlled conditions; (ii) representation-exchangeability and causal-sensitivity metrics expose failure modes that are not captured by plain accuracy; (iii) qualitative failure cases make the failure-mode taxonomy concrete; and (iv) exploratory LLaVA-family runs under the same zero-shot protocol remain below the main judge suite on this benchmark. We encourage researchers to use

Table 14: **Exploratory LLaVA-family model performance on MiniGrid Tier-0.** The metrics are analogous to those in Table 12 and are averaged over composite and split representations. These results are reported as auxiliary evidence rather than as part of the main model suite.

Model	Task C Acc.	Task A Acc.	Task E Acc.	Strict Rate
LLaVA-Critic-R1-7B	0.33	0.28	0.11	0.27
LLaVA-Critic-7B	0.65	0.25	0.00	0.00
LLaVA-OneVision-7B	0.63	0.27	0.00	0.00
LLaVA-OneVision-Chat	0.63	0.24	0.00	0.00
Mean	0.56	0.26	0.03	–

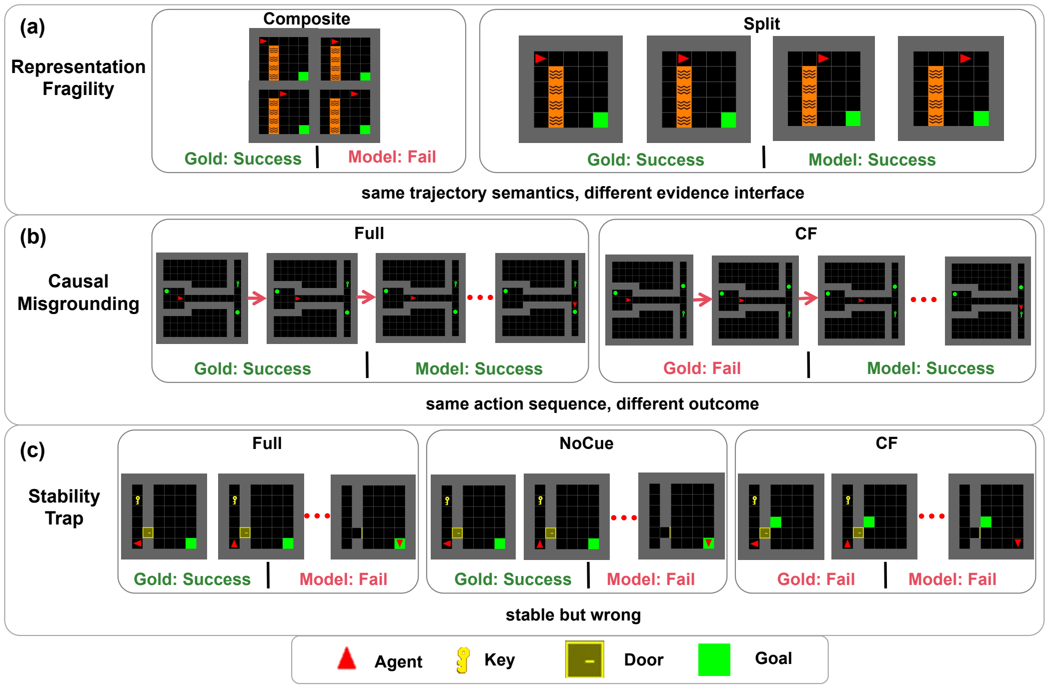


Figure 14: **Representative MiniGrid failure cases.** Each row shows the model-facing evidence, simulator-derived gold label, and model verdict before and after the diagnostic change. Row 1 illustrates representation fragility, where the same Full trajectory is judged differently under composite and split representations. Row 2 illustrates causal misgrounding, where a matched Full/CF pair receives an incorrect or unchanged verdict. Row 3 illustrates a stability trap, where the model remains stable across evidence conditions but is wrong or degenerate. Gold labels are derived from simulator state and are not shown to the model.

the machine-readable supplemental files for further analysis and to contribute additional models and probes to future versions of the benchmark.

E MiniWorld Extension

E.1 Environments and Tasks

We include a small MiniWorld extension as a 3D external-validity probe for GridWM-Judge. Unlike the Tier-0 MiniGrid suite, which is designed for tightly audited symbolic-state control, MiniWorld renders trajectories from a first-person 3D viewpoint. This introduces viewpoint variation, depth cues, partial occlusion, object-scale variation, and navigation ambiguity while preserving the same simulator-backed single-source-of-truth principle used throughout the benchmark. Figure 15 illustrates the MiniWorld evidence interface, audit-only simulator metadata, and matched success/failure structure.

The purpose of this extension is not to merge 3D results into the main MiniGrid leaderboard. Instead, MiniWorld tests whether the same diagnostic logic remains informative when the evidence interface becomes more realistic. For each MiniWorld rollout, we retain a simulator state trace, an action sequence, rendered visual evidence, a mission or task description, and an outcome label. The model-facing item contains only rendered evidence and task instructions; states, intervention records, labels, and validation metadata remain audit-only. Table 15 summarizes how the MiniWorld extension mirrors the SCI diagnostic logic while remaining separate from the Tier-0 MiniGrid results.

The MiniWorld extension focuses on three diagnostic questions: (i) whether the judge can use action-conditioned 3D evidence, (ii) whether success/failure judgments change under prompt framing,

Table 15: **MiniWorld extension design.** The 3D extension mirrors the SCI diagnostic logic of MiniGrid but is reported separately because the visual interface and trajectory distribution differ from the Tier-0 2D suite.

Component	Diagnostic role	Model-facing evidence	Reported output
MiniWorld suite	3D external-validity probe	First-person RGB storyboards; simulator state traces retained as audit metadata	Env./group/item counts
Task A-style	C2: action-conditioned evidence use	Current 3D observation, action text, and candidate next observations or trajectory slices	Acc_A
Task C-style	C1: presentation-sensitive trajectory judgment	Composite 3D storyboard with mission and action context	Acc_C
Framing probe	C1: prompt-framing sensitivity	Alternative framing prompt applied to the same 3D trajectory evidence	$\Delta_C^{\mathrm{frame}}$
Task D-style	C4: causal sensitivity	Successful trajectory evidence versus matched failure or counterfactual evidence	Acc_D, $\Delta_D^{\mathrm{frame}}$

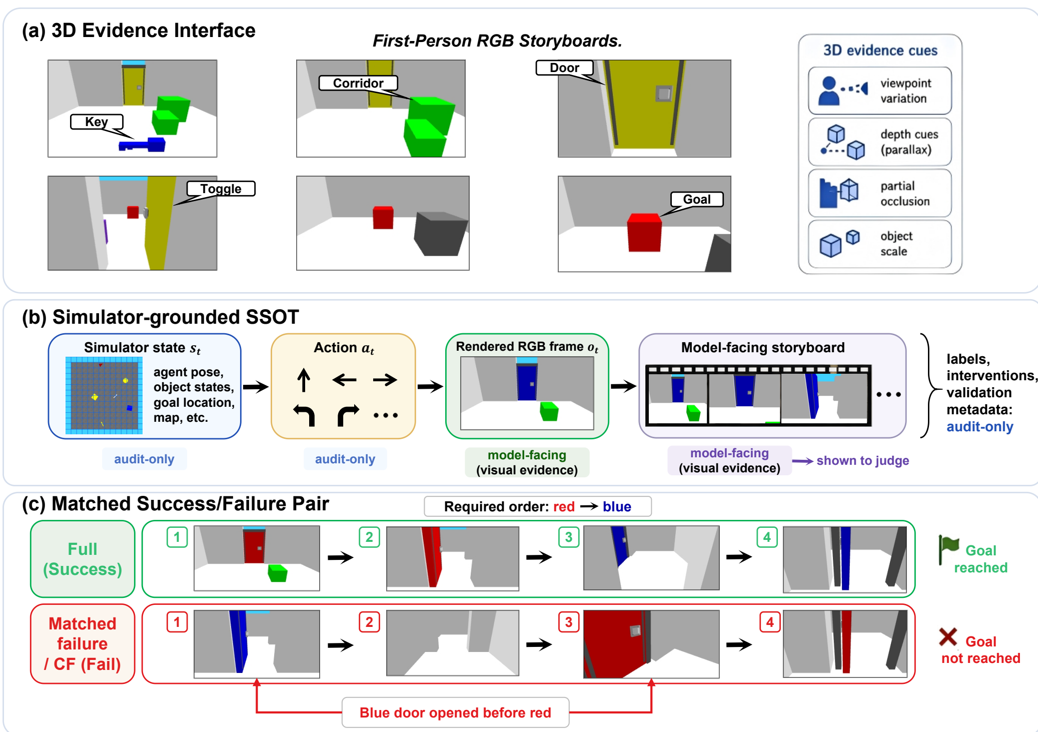


Figure 15: **MiniWorld evidence interface and simulator-grounded source of truth.** The figure shows representative first-person 3D frames, aligned simulator state/action traces, and a matched success/failure pair. Rendered frames are model-facing, while states, interventions, and labels remain audit-only metadata.

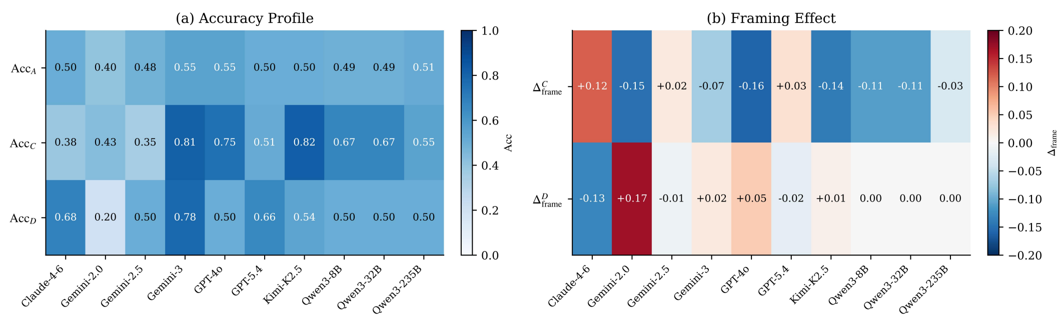
and (iii) whether the judge distinguishes successful trajectories from matched failure cases. We report the resulting accuracies separately from the MiniGrid Tier-0 results.

E.2 Results and Analysis

Table 16 reports MiniWorld results as a 3D external-validity probe rather than as a second leaderboard. The table should be read along three axes: action-conditioned evidence use (Acc_A), trajectory-level success/failure judgment (Acc_C), and Task-D-style failure discrimination (Acc_D). The framing-delta columns report how much the same evaluation changes under the finalized alternative framing prompt.

Table 16: **MiniWorld diagnostic results.** Values are point estimates from the MiniWorld composite-SSOT result file. Δ^{frame} denotes the change from the neutral prompt to the alternative framing probe; negative values indicate lower accuracy under framing. Because the current MiniWorld result file reports aggregate accuracies rather than paired Full–CF discrimination rates, Acc_D should be interpreted as a coarse Task-D-style failure-discrimination diagnostic rather than the full C4 IDR analysis used in MiniGrid. Support counts are provided in the released MiniWorld manifest.

Model	Acc_A	Acc_C	Δ_C^{frame}	Acc_D	Δ_D^{frame}
Claude Sonnet 4.6	0.50	0.38	+0.12	0.68	−0.13
Gemini 2.0 Flash	0.40	0.43	−0.15	0.20	+0.17
Gemini 2.5 Flash Lite	0.48	0.35	+0.02	0.50	−0.01
Gemini 3 Flash	0.55	0.81	−0.07	0.78	+0.02
GPT-4o	0.55	0.75	−0.16	0.50	+0.05
GPT-5.4	0.50	0.51	+0.03	0.66	−0.02
Kimi K2.5	0.50	0.82	−0.14	0.54	+0.01
Qwen3-8B	0.49	0.67	−0.11	0.50	+0.00
Qwen3-32B	0.49	0.67	−0.11	0.50	+0.00
Qwen3-235B	0.51	0.55	−0.03	0.50	+0.00
Mean	0.50	0.59	−0.06	0.54	+0.01



Model	Acc_A	Acc_C	Acc_D	Δ_C^{frame}	Δ_D^{frame}
Mean	0.50	0.59	0.54	-0.06	+0.01

Figure 16: **MiniWorld diagnostic profile across models.** (a) **Accuracy Profile**: point-estimate accuracies for three MiniWorld diagnostic axes: action-conditioned evidence use (Acc_A), trajectory-level success/failure judgment (Acc_C), and Task-D-style failure discrimination (Acc_D). Darker cells indicate higher accuracy, and the overlaid numbers give the corresponding point estimates. (b) **Framing Effect**: changes in accuracy under the finalized alternative framing prompt, computed relative to the corresponding neutral prompt as $\Delta_C^{\text{frame}} = \text{Acc}_C^{\text{frame}} - \text{Acc}_C$ and $\Delta_D^{\text{frame}} = \text{Acc}_D^{\text{frame}} - \text{Acc}_D$. Negative values indicate lower accuracy under framing, while positive values indicate higher accuracy. The summary row reports the mean value of each metric across models. The figure highlights that MiniWorld model rankings differ across diagnostic axes and that framing sensitivity is not reducible to overall accuracy.

Figure 16 visualizes the same MiniWorld results as a diagnostic profile, separating accuracy-style axes from framing effects across models. The main pattern is that no single MiniWorld accuracy number explains judge reliability. For example, GEMINI 3 FLASH is among the strongest models on the trajectory-level and Task-D-style columns, with $\text{Acc}_C = 0.81$ and $\text{Acc}_D = 0.78$. However, its Task-C score changes under framing ($\Delta_C^{\text{frame}} = -0.07$), showing that strong trajectory-level accuracy does not by itself imply presentation invariance. Similarly, GPT-4O obtains a high $\text{Acc}_C = 0.75$ but a lower $\text{Acc}_D = 0.50$, indicating that success/failure judgment and matched failure discrimination should be interpreted separately.

Model ranking also changes across diagnostic axes. KIMI K2.5 obtains the highest Acc_C in this MiniWorld evaluation (0.82), but its Task-D-style score is lower than that of GEMINI 3 FLASH (0.54 vs. 0.78). Conversely, CLAUDE-SONNET-4-6 performs better on Acc_D (0.68) than on Acc_C (0.38). This separation is consistent with the SCI framing: presentation robustness, trajectory-level correctness, and Task-D-style failure discrimination are distinct properties and should not be collapsed into a single aggregate score.

The mean MiniWorld scores remain close to the mid-range: $\mathrm{Acc}_A = 0.50$, $\mathrm{Acc}_C = 0.59$, and $\mathrm{Acc}_D = 0.54$. This suggests that the 3D extension is not saturated by current judges. In particular, several models remain near chance on Task-D-style metrics, and GEMINI-2.0-FLASH obtains a low $\mathrm{Acc}_D = 0.20$. Because the current MiniWorld result file reports aggregate accuracies rather than paired IDR-style discrimination rates, these numbers should be interpreted as coarse diagnostic evidence rather than as the full C4 analysis used in MiniGrid.

Compared with the 2D MiniGrid benchmark, the MiniWorld extension preserves the same methodological logic but exposes a different failure surface. Both settings use simulator-backed labels, aligned visual evidence, and controlled presentation or causal variants. MiniWorld additionally introduces first-person 3D perception burdens, including camera pose, partial observability, occlusion, depth cues, and object-scale variation. It is therefore best viewed as a stress test of whether SCI-style diagnostics remain informative under a richer evidence interface, not as a replacement for the fully audited Tier-0 MiniGrid suite.

F Implementation Details and Reproducibility

This appendix documents the implementation, data release, schema conventions, prompting and parsing protocol, and reproducibility information for GridWM-Judge. Appendix B describes how trajectories are generated and validated, Appendix C describes the model-facing task interfaces, and Appendix D reports supplementary results. Here we focus on the artifacts needed to inspect, rerun, and score the benchmark.

F.1 Code and Data Availability

The anonymized review artifacts are available at the following links:

Code: Anonymous GitHub repository.

Data: Anonymous Harvard Dataverse preview.

The released artifacts include the generated exam files, rendered images, ground-truth annotations, prompt templates, validation metadata, scoring scripts, and result summaries used in the paper. The code is released under the MIT License. The synthetic dataset artifacts are released under CC0 1.0 Universal unless otherwise stated in the repository metadata.

F.2 Repository Structure and Compute

The repository is organized around generation, inference, scoring, and export scripts. The top-level structure is:

```
GridWM-Judge/
  scripts/
    01_generators/minigrid/      # exam builders and per-task generators
    02_trajectories/             # MiniGrid and MiniWorld trajectory generation
    03_postprocess/framing/      # NoCue and CF variant construction
    04_inference/                # unified VLM inference gateway
    05_scoring/                  # scoring and metric aggregation
    06_rendering/                # figure and image rendering utilities
    08_export/                   # dataset export utilities
  outputs/                       # generated trajectories and exams
  config.py
  requirements_current.txt
  LICENSE
```

This work involves no model training or fine-tuning. Dataset generation and validation are CPU-bound and use fixed seeds. VLM inference is performed either through external APIs or local

Table 17: Main software dependencies used for generation, inference, rendering, and scoring.

Dependency	Version	License	Purpose
MiniGrid	3.0.0	Apache 2.0	Grid-world environments
Gymnasium	1.2.3	MIT	Environment interface
OpenAI SDK	2.15.0	Apache 2.0	API inference client
Transformers	4.57.3	Apache 2.0	Local VLM inference
PyTorch	2.8.0+cu128	BSD-style	GPU acceleration
Pygame	2.6.1	LGPL	Rendering support
Pillow	12.0.0	HPND	Image processing

inference for open-weight models. Local model runs use a single NVIDIA RTX 4060 Ti 16GB GPU. The MiniGrid data generation pipeline produces the paper-facing exam set in under two hours on a standard multi-core CPU server; the dominant runtime and monetary cost comes from VLM inference. Table 17 lists the main software dependencies used for generation, inference, rendering, and scoring.

F.3 Data Format and Metadata

The released benchmark uses JSONL files, with one exam item per line. Each item separates model-facing fields from audit-only fields. Model-facing fields are the only inputs provided to a VLM judge. Audit-only fields contain simulator states, labels, intervention metadata, validation logs, and other information used only for scoring and reproducibility.

A minimal item has the following logical structure:

```
{
  "uid": "C.keycorridor.g000001.full.orig.clean.comp",
  "task": "C",
  "schema_version": "gridwm.exam.v1",
  "env_task": "keycorridor",
  "group_id": "g000001",
  "variant": "full",
  "presentation": {
    "temporal": "orig",
    "visual": "clean",
    "framing": "neutral",
    "rep": "composite"
  },
  "model_input": {
    "images": ["relative/path/to/image_or_frame.png"],
    "prompt": "model-facing task instruction"
  },
  "audit_only": {
    "label": 1,
    "success": true,
    "state_seq": "relative/path/to/state_trace.json",
    "actions": ["forward", "toggle", "..."],
    "reward": 1.0,
    "nocue_meta": null,
    "cf_meta": null,
    "validator_status": "passed"
  }
}
```

Task-specific fields extend this common schema. Task A items additionally store the action condition and multiple-choice candidate order. Task B items store the structured perception target, including agent direction, carried object, front cell, and visible objects. Task C items store trajectory variant and presentation-probe metadata. Task D items store matched Full/CF group identifiers for causal-

Table 18: Model aliases used in the paper and corresponding inference route. Exact endpoint strings and run metadata are included in the released machine-readable logs.

Alias in paper	Model family / provider	Inference route
Claude Sonnet 4.6	Claude Sonnet family	API
GPT-4o	GPT-4o family	API
GPT-5.4	GPT-5-family judge run	API
Gemini 3 Flash	Gemini Flash family	API
Gemini 2.5 Flash Lite	Gemini Flash Lite family	API
Gemini 2.0 Flash	Gemini Flash family	API
Kimi K2.5	Kimi family	API
Qwen3-8B	Qwen open-weight family	local/API run
Qwen3-32B	Qwen open-weight family	local/API run
Qwen3-235B	Qwen large open-weight family	local/API run

sensitivity scoring. Task E items store the checkpoint index and object/status label. The complete schema files and example JSONL records are included with the released dataset.

For Task B, directions follow the MiniGrid convention:

Value	Direction
0	up
1	right
2	down
3	left

Door states follow the MiniGrid object-state convention: state 0 denotes open, state 1 denotes closed, and state 2 denotes locked. Object categories appearing in structured outputs include `key`, `ball`, `box`, `door`, `goal`, `wall`, `floor`, and `agent`; unseen or empty cells are not required in the model output unless a task explicitly asks for them.

F.4 Prompting, Decoding, and Parsing

All models are evaluated as external judges of model-facing evidence. The prompts contain only rendered images, mission or action text when required by the task, task instructions, and the expected answer format. They do not contain ground-truth labels, rewards, simulator states, validation flags, or intervention metadata.

The evaluated model aliases in the paper are summarized in Table 18. Exact provider endpoint strings, run timestamps, and API parameters are recorded in the released run metadata files.

When supported by the provider, decoding uses deterministic settings:

Parameter	Value
temperature	0.0
top_p	not set unless required by provider
top_k	not set unless required by provider
max tokens, Task A	512
max tokens, Task B	2048
max tokens, Task C/D/E	256

Invalid, empty, refused, or unparsable responses are counted as incorrect and logged. The parser is deterministic and task-specific:

Task A. The parser extracts the first isolated multiple-choice label in `{A,B,C,D}`. If no valid label is found, the response is marked invalid.

Tasks C and D. Binary success/failure judgments are parsed with case-insensitive keyword matching. Success words include `success`, `succeed`, `succeeded`, `successful`, and `yes`. Failure words

Table 19: Reproducibility summary.

Item	Documentation / release location
Code	anonymized GitHub repository
Dataset	anonymized Harvard Dataverse preview
Exam files	JSONL files in dataset release
Rendered images	image folders referenced by JSONL records
Prompt templates	repository and dataset metadata
Ground-truth labels	audit-only fields in JSONL records
Model outputs	prediction logs and score files
Scoring scripts	`scripts/05_scoring/`
Random seed	42
Training or fine-tuning	none
Dataset license	CC0 1.0 Universal
Code license	MIT License
Privacy	synthetic simulator data only; no human subjects or scraped images

include `fail`, `failed`, `failure`, and `no`. If both classes appear, the parser applies the deterministic conflict rule implemented in the released scoring script; otherwise invalid or ambiguous outputs are marked incorrect.

Tasks B and E. JSON outputs are parsed using strict JSON syntax. Task B requires the expected structured fields for the agent, front cell, and visible objects. Task E expects a JSON object with an `answer` field. Exact-match and component-level scores are both logged when applicable.

F.5 Reproduction Commands and Reporting

Table 19 summarizes the released artifacts and documentation needed to reproduce the benchmark results. A typical reproduction workflow is:

```
# 1. Install dependencies
pip install -r requirements_current.txt

# 2. Generate validated trajectories and exam items
python scripts/01_generators/minigrid/build_exam.py --seed 42

# 3. Run model inference
python scripts/04_inference/run_inference.py \
  --manifest outputs/exams/minigrid_manifest.jsonl \
  --model <model_alias>

# 4. Score predictions and export metrics
python scripts/05_scoring/score_exam.py \
  --predictions outputs/predictions/<model_alias>.jsonl \
  --manifest outputs/exams/minigrid_manifest.jsonl
```

The exact command wrappers and paths are provided in the repository README. The main reproducibility seed is 42 and is used for environment generation, trajectory planning, NoCue masking, and exam sampling when applicable.

Appendix A defines the statistical estimands and uncertainty-estimation procedures. Appendix D reports the supplementary MiniGrid results and qualitative failures. The released scoring scripts include the deterministic aggregation logic used to produce the tables and figures in the paper.

G Scope and Impact

This appendix clarifies the intended scope of GridWM-Judge, the limitations under which the main claims should be interpreted, and the broader impacts of releasing a simulator-grounded diagnostic benchmark for VLM judges. The purpose is not to make unrestricted claims about all VLMs or all embodied environments, but to make explicit what our controlled diagnostic evidence can and cannot support.

G.1 Scope and Limitations

Table 20 summarizes the scope of claims supported by the benchmark and the corresponding limitations.

Controlled diagnostic scope. GridWM-Judge is designed as a controlled diagnostic benchmark rather than a realistic robotics benchmark. The Tier-0 evaluation suite uses MiniGrid-style symbolic environments because they provide auditable states, actions, object identities, and success predicates. This control is essential for constructing aligned Full, NoCue, and CF variants, and for separating presentation changes from causal interventions. The corresponding limitation is that MiniGrid abstracts away continuous perception, 3D geometry, photorealistic appearance, multi-agent interaction, and real-world sensor noise. Our results therefore diagnose reliability failures under controlled embodied trajectories; they should not be interpreted as direct estimates of deployment performance in real robotic systems.

Environment and agent-behavior scope. The current benchmark emphasizes short-horizon, discrete-action navigation and interaction tasks with deterministic simulator dynamics. The trajectories are produced by deterministic planners so that the underlying success/failure labels and causal events can be audited exactly. This design does not cover noisy control, partial actuation failure, stochastic policies, exploration behavior, or suboptimal agent trajectories. Future versions should include more diverse behavior policies and longer-horizon tasks, but the present paper intentionally uses deterministic rollouts to isolate judge-side failures rather than agent-side failures.

Model and prompting scope. The evaluated models constitute a finite VLM-judge suite under the prompting and decoding protocol described in Appendix F. The benchmark is not a comprehensive leaderboard over all proprietary and open VLMs. The reported failures should be read as evidence that current representative judges can violate state-conditional invariance, not as a universal impossibility result. Likewise, we use standard judge prompts rather than systematically optimizing chain-of-thought prompting, few-shot demonstrations, tool use, self-consistency voting, or specialized fine-tuning. Better prompting or training may improve some scores, but such improvements would need to be tested under the same causal and presentation controls.

Task and SCI-condition scope. GridWM-Judge operationalizes four diagnostic conditions: presentation invariance, action-conditioned dynamics, representation exchangeability, and causal sensitivity. Task C primarily probes presentation invariance through temporal, visual, and prompt-framing perturbations; Task A probes action-conditioned transition use; the composite–split comparison probes representation exchangeability; and Task D probes causal sensitivity through matched Full/CF comparisons. Tasks B and E are local perception and state-tracking controls that help interpret higher-level failures. This coverage is diagnostic rather than exhaustive: a judge that performs well on these tasks may still fail under different environments, longer contexts, ambiguous goals, or naturalistic video evidence.

NoCue interpretation. NoCue removes selected early visual evidence while preserving the underlying successful trajectory. It is therefore an evidence-removal diagnostic, not a semantics-preserving presentation transformation in the same sense as temporal order, visual style, prompt framing, or representation format. We use NoCue to ask whether a judge relies on early causal cues, but we do not treat NoCue stability alone as proof of presentation invariance.

MiniWorld extension. The MiniWorld experiments in Appendix E are included as a 3D external-validity probe. They should be interpreted separately from the Tier-0 MiniGrid results because the 3D environments introduce different perceptual and rendering factors. We do not aggregate MiniWorld and MiniGrid results into a single headline score.

Threats to validity. We summarize three main validity threats. First, *internal validity* may be affected by parser ambiguity, API nondeterminism, or generator artifacts. We mitigate these risks

Table 20: Scope of claims and corresponding limitations.

Axis	What the paper supports	What remains outside scope
Domain	Controlled MiniGrid diagnostics with auditable states, actions, and success predicates.	Direct deployment claims for real robots, photorealistic video, or open-ended embodied tasks.
Agent behavior	Deterministic rollouts that isolate judge-side reliability failures.	Noisy control, stochastic policies, sub-optimal exploration, and long-horizon recovery behavior.
Presentation	Temporal, visual, prompt-framing, and representation-format probes over matched trajectory semantics.	All possible UI formats, natural-language task framings, or video encodings.
Models	A finite VLM-judge suite under a fixed evaluation protocol.	Universal claims about all VLMs, future models, or heavily optimized prompting/training regimes.
Metrics	Diagnostic evidence for correctness, presentation stability, representation exchangeability, and causal sensitivity.	A single deployment-readiness score or safety certificate.

by using constrained answer formats, deterministic parsing rules, fixed seeds where applicable, and validation gates over generated trajectories. Second, *external validity* is limited by the synthetic grid-world domain and by the finite set of evaluated models. The MiniWorld extension partially probes this issue, but does not eliminate it. Third, *construct validity* depends on whether SCI captures the aspects of judge reliability that matter in downstream use. We view SCI as a necessary diagnostic principle: a reliable judge should not change its verdict merely because evidence is reformatted, and should change when a causal intervention changes the outcome. However, SCI is not sufficient for deployment safety, because real settings may involve ambiguous goals, incomplete observations, distribution shift, and human factors not represented here.

G.2 Societal Impact and Outlook

Table 21 summarizes the main impact considerations and recommended mitigations for responsible release and use.

Positive impacts. VLM judges are increasingly used to evaluate multimodal and embodied systems. A diagnostic benchmark such as GridWM-Judge can help researchers identify when a judge is correct for the wrong reason, unstable under benign presentation changes, or insensitive to causal outcome changes. This may reduce overreliance on opaque aggregate accuracy and encourage more careful reporting of judge reliability, especially in evaluation pipelines that influence model selection, dataset curation, or simulated safety testing.

Risks of overclaiming and misuse. The main risk is not that the benchmark directly enables harmful physical actions, since it uses synthetic grid-world data and releases no personal, private, or safety-critical operational data. The more realistic risk is *overclaiming*: users may treat high GridWM-Judge scores as a guarantee that a VLM judge is safe for real-world robotics or high-stakes decision-making. This would be inappropriate. The benchmark should be used as one diagnostic layer, not as a deployment certificate. Another risk is benchmark gaming: once examples and prompts are public, models or systems may be tuned to exploit the benchmark without genuinely improving causal grounding. Future releases should therefore include held-out diagnostic slices and versioned evaluation protocols.

Responsible release. The dataset is synthetic and simulator-generated. It does not contain human-subject data, personal information, scraped images, or private trajectories. The release should nevertheless include documentation of generation scripts, prompt templates, schemas, validation logs, licenses for dependencies, and known limitations. We recommend that any public release clearly state that the benchmark evaluates VLM judges under controlled MiniGrid and MiniWorld conditions, and that it is not intended as a standalone safety evaluation for deployed embodied systems.

Table 21: Impact considerations and recommended mitigations.

Consideration	Potential concern	Mitigation in this paper / release
Overclaiming	Treating SCI scores as deployment-readiness guarantees.	Explicitly state the controlled diagnostic scope and report limitations alongside results.
Benchmark gaming	Tuning models to public prompts or examples without improving real reliability.	Use versioned splits, document probes, and encourage held-out extensions.
Misuse of judge diagnostics	Crafting inputs that exploit known judge weaknesses.	Frame results as reliability audits and avoid presenting the benchmark as an adversarial attack toolkit.
Privacy and data rights	Releasing sensitive images or trajectories.	Use simulator-generated synthetic data with no human subjects or private data.
Compute cost	Repeated VLM inference across many variants.	Provide modular subsets, deterministic scripts, and transparent compute reporting.

Environmental considerations. MiniGrid trajectory generation and validation are lightweight relative to large-scale VLM inference. The main compute footprint comes from repeated model calls over multiple tasks, variants, and presentation probes. The benchmark is intentionally modular so that researchers can run smaller diagnostic subsets when compute is limited. Reporting compute resources and inference settings in Appendix F is important for reproducibility and for understanding the environmental cost of evaluation.

Future directions. The most important next step is to test whether the same SCI failures persist in richer domains, such as longer MiniWorld tasks, photorealistic simulators, real robot videos, and embodied tasks with ambiguous or compositional goals. A second direction is to expand the model suite, including open-weight VLMs, video-language models, and specialized judge models. A third direction is to evaluate mitigation strategies such as structured output constraints, calibrated abstention, evidence localization, multi-view inputs, self-consistency, or training on counterfactual diagnostic data. Future benchmark versions should also include adversarially selected but simulator-valid trajectories to test whether apparent judge reliability survives more challenging causal and presentation shifts.

Conclusion on scope. GridWM-Judge provides controlled evidence that VLM judges can violate state-conditional invariance through presentation sensitivity, representation fragility, action-insensitive heuristics, causal misgrounding, and degenerate stability. These conclusions are bounded by the benchmark domain, evaluated model suite, and prompting protocol. Within those bounds, the benchmark offers a reproducible diagnostic framework for asking whether a VLM judge changes its verdict for the right reason, but it should not be treated as a standalone deployment or safety certificate.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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   Justification: The abstract and introduction state the SCI principle, the GridWM-Judge benchmark, the controlled MiniGrid/MiniWorld diagnostic scope, and the main empirical findings. The scope of the claims is tied to the benchmark design in Sections 1, 3, and 4, and limitations are summarized in Appendix G.1.

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Appendix G.1 explicitly discusses the controlled diagnostic scope, synthetic grid-world domain, deterministic rollouts, finite model suite, prompting scope, MiniWorld extension, and threats to validity. The paper also states that GridWM-Judge is a diagnostic benchmark rather than a deployment-readiness certificate.

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   Answer: [Yes]

   Justification: No model training is performed; the experiments are controlled evaluations of VLM judges. The paper specifies the evaluated tasks and environments, model-facing evidence, inference protocol, deterministic decoding when supported, invalid-output handling, metrics, prompts, and parsing rules in Section 5.1 and Appendix F.

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Answer: [Yes]

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